

Article

Land Use Change Impacts over the Indus Delta: A Case Study of Sindh Province, Pakistan

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Abstract: Land use and land cover changes (LULCCs) are vital indicators for assessing the dynamic relationship between humans and nature, particularly in diverse and evolving landscapes. This study employs remote sensing (RS) data and machine learning algorithms (MLAs) to investigate LULCC dynamics within the Indus River Delta region of Sindh, Pakistan. The focus is on tracking the trajectories of land use changes within mangrove forests and associated ecosystem services over twenty years. Our findings reveal a modest improvement in mangrove forest cover in specific areas, with an increase from 0.28% to 0.4%, alongside a slight expansion of wetland areas from 2.95% to 3.19%. However, significant increases in cropland, increasing from 22.76% to 28.14%, and built-up areas, increasing from 0.71% to 1.66%, pose risks such as altered sedimentation and runoff patterns as well as habitat degradation. Additionally, decreases in barren land from 57.10% to 52.7% and a reduction in rangeland from 16.16% to 13.92% indicate intensified land use conversion and logging activities. This study highlights the vulnerability of mangrove ecosystems in the Indus Delta to agricultural expansion, urbanization, resource exploitation, and land mismanagement. Recommendations include harmonizing developmental ambitions with ecological conservation, prioritizing integrated coastal area management, reinforcing mangrove protection measures, and implementing sustainable land use planning practices. These actions are essential for ensuring the long-term sustainability of the region's ecosystems and human communities.

Keywords: coastal areas; Indus Delta; land use land cover change; machine learning algorithms; mangroves



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1. Introduction

In an era marked by rapid urbanization, agricultural intensification, and climate change, understanding the intricate dynamics of land use and land cover (LULC) is critical. Examining these dynamic shifts is fundamental for promoting sustainable land management practices, preserving biodiversity, and shaping informed environmental policies [1–3]. Over the past two decades, Pakistan has experienced significant LULC changes characterized by rapid urbanization and agricultural intensification [4,5]. These changes have led to the degradation of natural ecosystems and intensified pressure on water resources, exacerbating issues like urban heat islands (UHIs) in major cities such as Karachi, Lahore, and Faisalabad [6,7]. Furthermore, these alterations have heightened susceptibility to floods and droughts [8,9].

The impact of LULC extends beyond environmental consequences, especially in Karachi, Pakistan's largest and most economically significant city. Urbanization in Karachi has reshaped the socio-economic landscape, affecting livelihoods dependent on natural

resources [10]. In landscapes as diverse as the Indus Delta in Pakistan, where urban, agricultural, and natural ecosystems intersect, studying LULC becomes imperative [11–14]. Significant land use and land cover changes (LULCCs) have been observed in Sindh Province over the past two decades, driven primarily by increased urbanization and agricultural expansions aimed at meeting food requirements amidst climate change [15,16].

Among these transformations, the Indus Delta, housing one of the world's most biologically rich mangrove ecosystems, faces significant environmental challenges, primarily due to upstream freshwater extraction for agriculture and urbanization [17,18]. Mangroves, resilient woody vegetation in tropical and subtropical coastal regions worldwide, play a central role in marine and terrestrial ecosystems, contributing significantly to environmental health, biodiversity, and the provision of essential resources such as food, fuel, and feed. Despite their importance, mangroves face substantial threats from various anthropogenic activities, particularly marine pollution. These salt-tolerant woody plants shape delicate coastal ecosystems at the interface of land and water, with *Avicennia marina* dominating the mangrove forest, alongside species like *Aegiceras corniculatum*, *Ceriops tagal*, and *Rhizophora mucronata* [19].

In the early stages of digital mangrove mapping using remote sensing (RS) data, supervised image classification methods, including random forest (RF), support vector machine (SVM), and maximum likelihood classifier (MLC), were commonly employed to determine mangrove size and composition based on satellite images [20–23]. However, these studies often lacked comprehensive spatial and temporal analyses and did not fully integrate advanced machine learning algorithms (MLAs) with RS technologies. Furthermore, there has been insufficient integration of socio-economic factors and their interactions with environmental changes, leading to a limited understanding of the highest impacts of LULCC [9]. Pakistan, wherein mangroves cover approximately 600,000 hectares, ranks as the third-highest country globally in this regard [24]. The mangrove forest in the Indus Delta extends from Korangi to the Indian border at Sir Creek, harboring a unique plant community and significant biological value [24,25]. However, these forests have faced extensive threats from changes in environmental conditions and human activities over the past few decades. Land use changes, including urban sprawl and agricultural practices, have led to the reduction of mangrove areas. Additionally, water diversions, pollution, and over-exploitation have contributed to the deterioration and degradation of mangrove ecosystems in the Indus Delta.

This study addresses these gaps by utilizing a strong methodological framework that combines Landsat data with MLAs to provide a detailed analysis of LULCC dynamics over three decades. Focusing on the Indus Delta, this research highlights the interplay between urbanization, agricultural expansion, and mangrove ecosystem changes, offering new insights into the region's environmental challenges and conservation needs. The innovative use of MLAs, such as RF and SVM, in conjunction with RS data, ensures high accuracy in LULCC mapping and analysis. The primary objectives of this study are to quantify the extent and nature of LULCCs in the Indus Delta from 1990 to 2020, identify key drivers and patterns, assess the implications of these changes for ecosystem services and biodiversity with a particular focus on mangrove conservation, and inform sustainable land management practices and policy interventions by providing actionable insights based on complete spatial and temporal data.

By employing advanced geospatial analysis, historical trend analysis, and prediction modeling, this study ensures a robust methodological approach and technical consistency. Leveraging RS technology, which has revolutionized environmental monitoring by enabling periodic and systematic observations of large-scale landscapes, generates extensive datasets that are vital for LULCC studies [26,27]. Aligned with geospatial techniques, RS data provide essential spatial and temporal information for analyzing LULCC patterns [28]. Recent advancements in RS technology have introduced new techniques such as image differencing, ratio, and principal component analysis (PCA) methods for monitoring LULCCs [29,30]. Landsat data and MLAs, particularly RF and SVM, have emerged as

cost-effective and accurate methods for data collection and analysis over large regions, surpassing traditional field survey techniques [31–34].

The advanced use of MLAs, geospatial analysis, historical trend analysis, and prediction modeling in this study provides detailed facts and figures of LULCC dynamics in the Indus Delta. Drawing on recent scientific methods and techniques, such as advanced space-borne thermal emission and reflection radiometer (ASTER) [35], landscape pattern indices in response to LULCCs [36], and climate change mitigation through afforestation and reforestation [37], this study ensures a robust methodology and technical consistency. The previous literature, including studies on change detection and computer processing of RS images [38], mapping global forest cover change using RS [39,40], and utilizing RS data and MLAs [41,42], further enrich the analytical framework of this study.

Furthermore, this research highlights the significant biodiversity and ecosystem health impacts due to land cover changes in diverse ecosystems such as arid deserts, fertile plains, and coastal regions. Effective conservation strategies are necessary to mitigate these impacts. As a key agricultural area, urban expansion and deforestation threaten productivity, making future change predictions crucial for sustainable practices and food security. Land cover changes also influence climate change by altering carbon sequestration and greenhouse gas emissions, underscoring the need for mitigation and adaptation strategies. These changes affect water availability and quality, guiding sustainable water resource management in this water-scarce region. Rapid urbanization requires careful planning to balance development with environmental sustainability, and predictive models can aid in minimizing negative impacts. Understanding land cover changes' effects on natural disaster frequency and severity is vital for disaster risk reduction.

Core attention has been given in this study to the dynamics of LULCCs in the Indus Delta of Pakistan from 1990 to 2020. It employs advanced RS techniques and MLAs to elucidate the drivers and patterns of LULCC, focusing on urbanization, agricultural expansion, and their impacts on mangrove ecosystems. The objectives include understanding the extent and nature of LULCCs, assessing the implications for ecosystem services and biodiversity, and informing sustainable land management practices and policy interventions. Research on land cover changes in Sindh, Pakistan, faces several challenges and includes numerous study gaps. High-resolution time-series data gaps affect the accuracy of machine learning models. Socio-economic factors must be integrated for a comprehensive understanding of changes, but data availability is limited. Different MLAs have varied effectiveness, necessitating careful selection. The interdisciplinary nature of LULC requires approaches that combine ecology, geography, and economics while predicting future changes involves dealing with significant uncertainties. By addressing these objectives, this study contributes to the broader understanding of environmental change in the Indus Delta region and highlights the importance of balancing socio-economic development with environmental conservation efforts.

By addressing these objectives, this study contributes to the broader awareness of environmental change in the Indus Delta region and highlights the importance of balancing socio-economic development with environmental conservation efforts. Additionally, this study emphasizes the need for effective conservation strategies to mitigate the impacts of LULCC on biodiversity and ecosystem health, ensuring sustainable development in one of Pakistan's most ecologically and economically vital regions.

2. Materials and Methods

2.1. Study Area

The study area, Sindh Province in Pakistan, encompasses geographical coordinates approximately ranging from 23.68° N to 28.42° N latitude and from 66.64° E to 71.02° E longitude. It is situated in the southern part of Pakistan (Figure 1), bordering the Arabian Sea to the south and sharing boundaries with the provinces of Balochistan to the west and Punjab to the north. Sindh Province boasts a diverse landscape characterized by arid plains, fertile river valleys, and coastal mangrove forests. The mighty Indus River, one of the

longest rivers in the world, flows through the heart of the province, shaping its geography and providing vital water resources for agriculture and human consumption. The coastal region of Sindh, particularly along the Arabian Sea, is home to the Indus Delta, one of the largest arid climate mangrove forests in the world. This unique ecosystem supports a rich biodiversity of flora and fauna, serving as a habitat for numerous species of migratory birds, marine life, and mangrove vegetation. The research conducted in Sindh Province focuses on understanding the dynamics of LULC within this diverse landscape. It examines how human activities, such as urbanization, agricultural expansion, and industrialization, are impacting the region's ecosystems, biodiversity, and human livelihoods. By analyzing LULC changes in Sindh Province, researchers aim to identify key drivers of change and their implications for ecosystem services. This knowledge can inform decision-making processes related to sustainable land management, conservation efforts, and socio-economic development in the region.

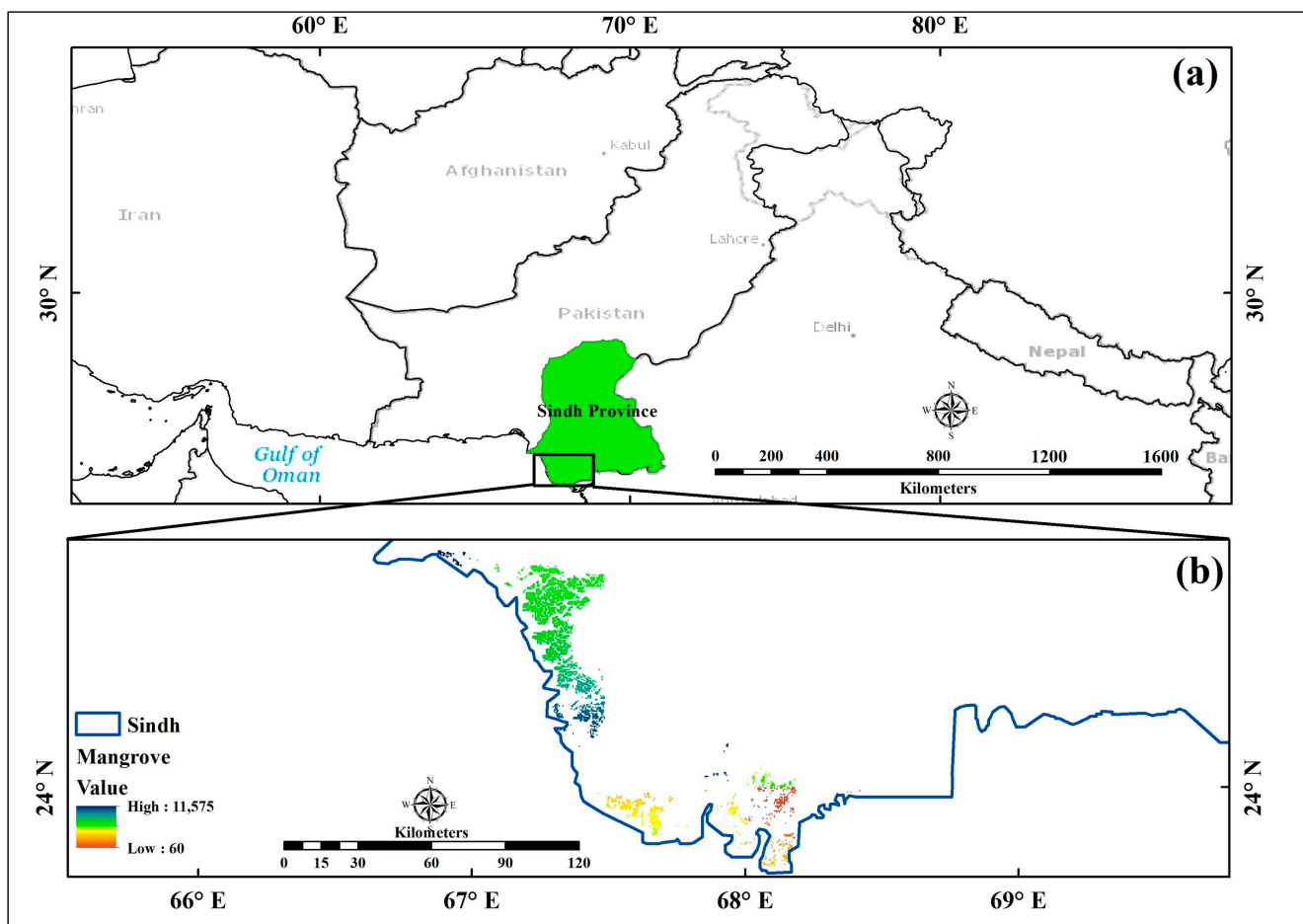


Figure 1. Map (a) shows the study area in Sindh province, Pakistan, and (b) shows the mangrove area in Sindh.

2.2. Data Collection and Preprocessing

In this study, the data collection was primarily focused on utilizing remote sensing (RS) data to capture the spatial and temporal dynamics of LULCCs in the Indus Delta region of Sindh province. To achieve this, multi-temporal Landsat imagery was acquired, providing a comprehensive historical perspective on land cover changes spanning two decades, a similar approach to that applied by [43]. Landsat data, renowned for their moderate resolution of 30×30 m, prove to be highly effective at capturing large-scale land cover changes. This characteristic is important in coastal area studies, where significant alterations in land use patterns are frequent [44].

This study also integrates field surveys, which serve as a crucial component for validating and strengthening the multi-spectral satellite imagery. Field surveys provide essential ground truth data, facilitating accurate land cover classification and analysis [45,46]. These surveys were specifically designed to identify key land use types that are prevalent in coastal Sindh, such as urban areas, agricultural land, and mangrove forests. A total of 540 samples were obtained from field surveys, with 400 being used for classification and 140 being used for verification.

Additionally, various other datasets were collected, including topographic maps, demographic data, and climate variables such as precipitation and temperature. These supplementary data sources support the contextual information, aiding in the interpretation of satellite data, particularly in understanding the relationships between land cover changes and socio-economic factors [46]. Climate data were sourced from WorldClim (<https://www.worldclim.org/data/annual.html>, accessed on 15 April 2024), while population and socio-economic indicators were obtained from the Pakistan Bureau of Statistics website (<https://www.pbs.gov.pk>).

While the core methodology aligns with recent studies [36,42,47,48], significant emphasis was placed on adapting these data collection methods to the unique geographical features and challenges of the coastal regions. This included a focused approach towards capturing the dynamics of LULCC and unique vegetation types such as mangroves. To minimize the impact of seasonal variations, data collection was conducted from March to April. Additionally, severe criteria were applied to ensure that cloud coverage in all Landsat images was below 10%. Remarkably, cloud cover was found to be consistently below 10% in all images examined within the study area (Table 1). Further metadata about these images were gathered from the USGS repository.

Table 1. Detail of all Landsat images with their sources.

Acquired Date	Spacecraft ID	Sensor ID	Path/Row	No. of Scene
2000	Landsat-5	TM	150-52/40-44	13–14
2005	Landsat-5	TM	150-52/40-44	13–14
2010	Landsat-5	TM	150-52/40-44	13–14
2015	Landsat-8	OLI_TIRS	150-52/40-44	13
2020	Landsat-8	OLI_TIRS	150-52/40-44	13

2.3. Land Use and Land Cover Classification and Analysis

This stage involves the application of various classification techniques to the pre-processed satellite data, which is aimed at categorizing land into different cover types. Given the distinctive characteristics of coastal landscapes, the selection and adaptation of classification methods were carefully followed in recent studies [41,46,49–52]. Supervised classification techniques involved training algorithms using known data points (training samples) obtained from field surveys. These algorithms were then applied to classify the entire dataset [51]. Special attention was paid to selecting training samples that represented a diverse range of land cover types, including urban areas, agricultural lands, and mangroves. Additionally, unsupervised classification methods were employed to complement the supervised techniques [52]. These methods clustered the data based on spectral properties without prior knowledge of land cover types, proving particularly valuable in identifying unique land cover classes specific to coastal areas, such as various types of wetlands or sandbanks [53].

The choice of classification algorithms has a key role in ensuring accurate results. Algorithms like the MLC and RFs were preferred due to their robustness and ability to handle complex data structures [54]. These algorithms were fine-tuned to account for the specific spectral characteristics of coastal regions, such as high reflectance from sandy areas and low reflectance from water bodies. The adoption of the MLC technique was influenced by its straightforward method for sample collection and its proven track record of yielding accurate results, as corroborated by previous research [48,55,56]. The

false-color combination of near-infrared, red, and green bandwidths was utilized for training sample collection. A stratified random sampling technique was adopted, and subsequently, a signature file was generated. These signature files for each image (spanning from 2000 to 2020 with a five-year temporal span) were utilized for MLC in ArcGIS 10.8.3 software. Area calculation and other statistical analyses, including an area change analysis using Microsoft® Excel® for Microsoft 365 MSO (Version 2311 Build 16.0.17029.20028) 64-bit and MATLAB R2024a, were conducted.

Land Use Land Cover Change and Its Indices

In addition to classification, this phase involved a meticulous analysis of the classified images. This included assessing changes in land cover over time and understanding the spatial patterns of these changes. Techniques such as change detection algorithms and spatial analysis tools were employed to elucidate the complex dynamics of land cover changes in the coastal environment. To ensure an accurate LULCC analysis, spatial indices such as the normalized difference water index (NDWI) for mangrove identification, the normalized difference vegetation index (NDVI) for all types of vegetation cover identification, and the normalized difference built-up index (NDBI) for precise built-up identification were incorporated [57–59].

2.4. Accuracy Assessment of the Classified Images

The accuracy assessment phase is important for any LULC studies. To ensure a comprehensive representation of all LULC types in the Indus Delta, approximately 500 sample locations were meticulously selected using the stratified sampling tool in ArcGIS 10.8.3 software, specifically the “Accuracy Assessment” tool under the spatial analysis environment. These samples served as ground truthing points to validate all classified maps [48]. High-resolution imagery from Google Earth Pro and the Global Positioning System (GPS) were employed in 2020 to collect these samples, ensuring a current and precise reflection of ground conditions for LULCC analysis. For the year 2005, in the absence of ground truthing data, the *Land Cover Atlas of Pakistan* (Sindh Province) series was utilized due to the lower resolution of Google Earth’s image data before 2005. This atlas was prepared through the collaborative efforts of The Space and Upper Atmosphere Research Commission (SUPARCO) and the Food and Agricultural Organization (FAO) [60].

The accuracy of the classified maps was evaluated through the cross-examination of the collected ground truth samples with the classified maps. A producer’s accuracy (PA) was utilized to determine the percentage accuracy of the forecasts provided for a class, while user accuracy (UA) was calculated by analyzing the referenced data of a class and expressed as a percentage. For further detailed map-based accuracy measures, [61] provides additional information and a methodology. To assess the accuracy of land cover classifications, error matrices (confusion matrices) were employed to compare the classified data with ground truth data obtained from field surveys. These accuracy assessment steps offer essential metrics such as overall accuracy, user accuracy, and producer accuracy, providing a quantitative evaluation of the classification results. Additionally, kappa statistics are utilized to offer a more robust evaluation of classification accuracy than overall accuracy alone [52,62,63].

2.5. Predictor Driver Extraction

2.5.1. The Role of Elevation and Slope

Elevation stands as a significant environmental factor influencing LULCC dynamics, such as agricultural practices, economic activities, and alterations in wetland areas. The elevation of the study area was precisely calculated using the 30×30 m digital elevation model (DEM). In this study, the DEM utilized is derived from the Shuttle Radar Topography Mission (SRTM) dataset, known for its global coverage and moderate resolution. Additionally, the slope gradient of the research area was derived from the DEM in ArcGIS 10.8.3 using the “Slope” tool within the 3D Analyst tool window [48]. Slope analysis

provides critical insights into terrain characteristics that influence land use patterns and ecosystem dynamics.

2.5.2. Evaluating the Distance from the River and Roads

Calculating the distance from the river involved a two-step process. Firstly, the watershed of the research area was delineated from the DEM using the ArcHydro tool extension in ArcGIS 10.8.3 software [64]. Subsequently, after delineating the watershed, the distance from the river was computed using the “Euclidean Distance” tool within the ArcGIS spatial analyst environment using a similar approach as that used in the study [48].

Similarly, determining the distance from roads also involved a two-step procedure. Initially, all road data obtained from OpenStreetMap (OSM) (<https://www.openstreetmap.org>) were consolidated into a single file. Subsequently, the utilization of the “Euclidean Distance” tool in ArcGIS 10.8.3 software facilitated the calculation of distances from roads.

2.5.3. Understanding Precipitation and Temperature Dynamics

Precipitation is a fundamental component of climate, exerting a profound influence on both the environment and human activities. Thus, access to reliable rainfall data is dominant for predicting shifts in land use and for implementing effective environmental and agricultural strategies. To this end, average annual precipitation data spanning the previous two decades of 2000–2022 was procured from the WorldClim website (<https://worldclim.org/data/monthlywth.html>, accessed on 15 April 2024). Subsequently, meticulous pre- and post-processing procedures of the data were conducted using ArcGIS 10.8.3 software, culminating in the generation of appropriate maps. This dataset serves as a crucial input for future predictions and simulations about LULCC.

Temperature is the main parameter in the transformation of LULC, exerting significant influence on human activities and ecosystems alike. It impacts biodiversity, plant life cycles, and the distribution of vegetation, leading to shifts in climate zones and triggering adaptive responses from wildlife as temperatures fluctuate. Average annual temperature data spanning the last two decades of 2000–2022 were sourced from the WorldClim portal. Similar to precipitation data, rigorous pre- and post-processing procedures were undertaken to develop pertinent maps.

2.6. Future Land Cover Simulations and Projections

In this section, advanced spatial modeling techniques combining GIS and RS methodologies play an important role in accurately predicting future LULC scenarios [65]. Notably, advanced models like the multi-layer perceptron–Markov chain (MLP-MC) were employed, integrating tools such as the MOLUSCE plugin for QGIS and the cellular automata (CA) model. These models, particularly through the control of the MLP-MC framework, support forecasting LULC transformations, capturing both static and dynamic elements to inform land use planning and sustainable strategies [66–68]. To predict LULCC, the model integrates various variables, including historical LULC alterations derived from Landsat imagery for the years 2000 and 2020, as well as independent variables such as the distance (buffer) to key features like water bodies and transportation networks. Additionally, climatic predictive variables such as annual average precipitation and temperature were incorporated into the modeling process. To ensure compatibility among DEM, Landsat images, and climatic variables, a resampling technique was employed to achieve spatial resolution consistency through upscaling. Validation of the model’s precision was conducted using the QGIS-MOLUSCE validation module, assessing kappa coefficients and percent correctness for classified and projected LULC maps for 2020. Once satisfactory accuracy levels were attained, the model was applied to predict LULC dynamics for the years 2025 and 2030.

3. Results

3.1. Land Use Land Cover Classification

This research utilized a total of five stacked preprocessed Landsat images to analyze LULCCs over the past two decades. Landsat-5 TM imagery from the year 2000 served as the base layer, while the most recent Landsat-8 OLI image from 2020 was selected as the final layer to detect changes in land cover. To enable interpretation, the temporal span was divided into three orders: from 2000 to 2010, from 2010 to 2020, and a predictive phase for the future up to 2030. Furthermore, (Table 2) shows the LULC classes in detail. The individual results of the five LULC classifications are presented in the following sections.

Table 2. Detailed description of all classified LULC types.

Class Code	LULC Type	Description
1	Forest	Forest, natural and artificial dense vegetation area, orchard
2	Wetland	Ocean and surface water bodies like major and minor streams, lakes, mangroves, and ponds
3	Cropland	All types of cultivated land including rainfed agriculture land
4	Built-up	Artificial structures and surfaces associated with urban and suburban environments
5	Barren land	Open spaces with low and no vegetation, deforested areas, rock surfaces, sand, and soil deposits
6	Rangeland	Grasslands, shrublands, savannas, and woodlands

In the year 2000, forest cover accounted for only 0.3% of the land area, while wetlands comprised 3%. Cropland covered 22.8% of the area, primarily along the river banks. Barren lands, occupying (57.1%) of the research area, were predominant on both the eastern and western margins of the province (Figure 2a) in the classified LULC map of 2005. By 2005, the distribution remained largely unchanged, with cropland and wetland areas increasing slightly at the expense of barren land and rangeland. The expansion of cropland, particularly evident in the northern Shikarpur and Sukkur areas and the southwestern regions around Keenjhar lake, indicates intensified agricultural activity.

Significant changes occurred in LULC by 2010, primarily due to disastrous floods. Forest areas decreased slightly, while wetlands expanded. Cropland covered the largest proportion (27.2%) of the land, with built-up areas accounting for only 1%. Barren lands still stand out in half (51.7%) of the research area (Figure 2c). Rangeland also experienced a slight increase. Notable changes in land cover were observed in the northwestern region, including in Jacobabad.

The 2015 LULC map exhibited similarities to the previous period, with a slight decrease in cropland and an increase in barren land. Barren land remained the dominant cover (54.4%), followed by cropland (26.6%) and rangeland (14.3%). Significant changes occurred in Sukkur, Badin, and Thatta, particularly in barren land distribution. By 2020, barren land continued to lead the study area at (52.7%), followed by cropland (28.1%), wetlands (3.2%), and built-up areas (1.7%). Forest cover remained minimal at 0.4% of the land area.

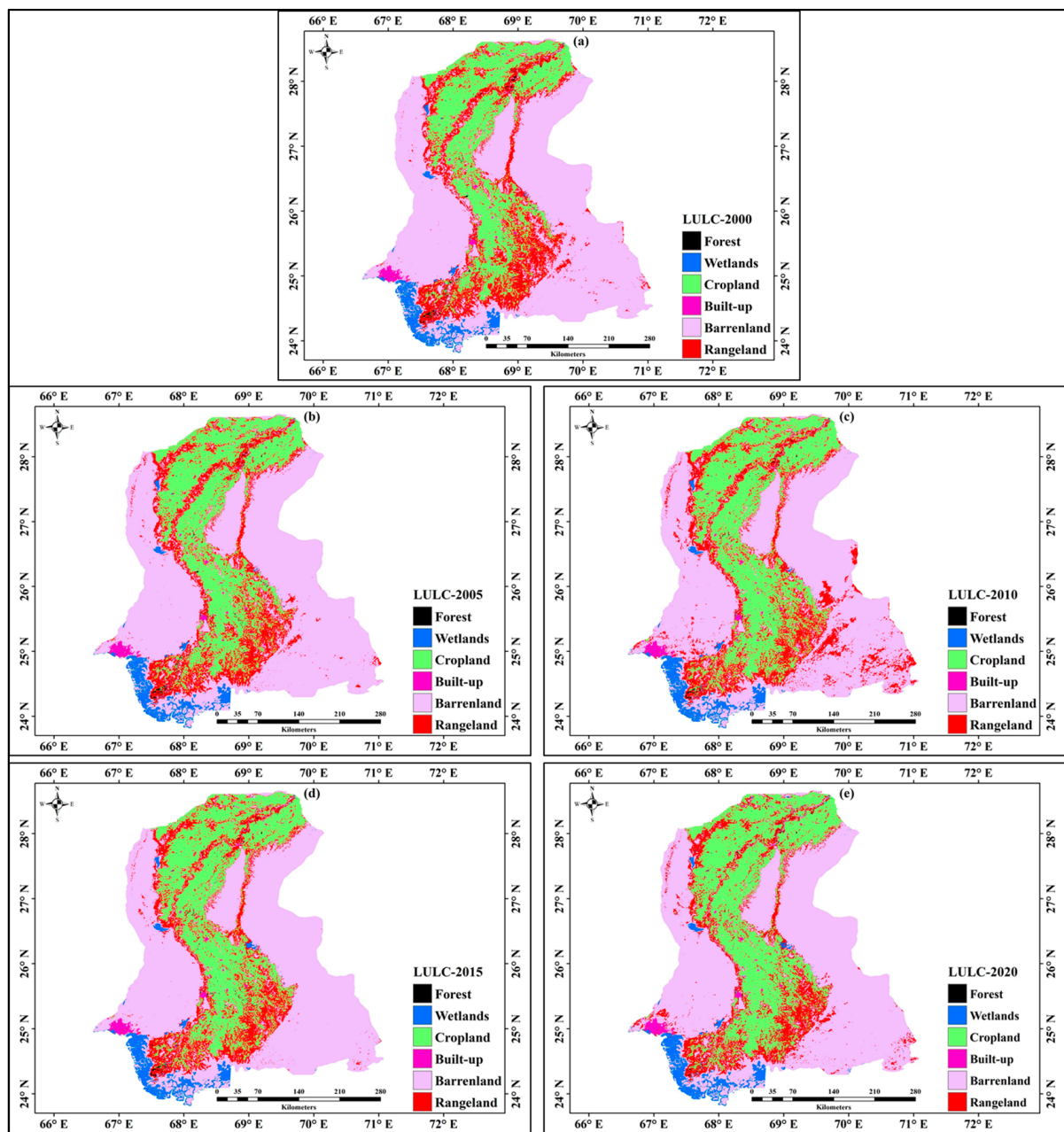


Figure 2. (a–e) LULC maps of the Indus Delta from 2000 to 2020.

3.2. Extraction of Indices

LULC indices have a significant role in forecasting LULCC over time, with some experts arguing for their superior precision and reliability compared with classified LULC maps. In this study, three key indices were identified: NDVI, NDBI, and NDWI. Both the NDVI and NDBI produce values ranging from -1 to $+1$, but in opposite directions. For example, NDVI values closer to $+1$ indicate areas with dense vegetation, while NDBI values closer to -1 indicate open water or bare land.

Threshold values were applied to the NDVI results, categorizing them into three classes: water, areas devoid of vegetation, and areas with varying types of vegetation. Specifically, wetlands, which typically include water bodies and vegetation, were identified using a combination of NDVI thresholds to distinguish between water and vegetation cover. For the base year (2000), the NDVI analysis indicated that 57.8% of the study area consisted of no-vegetation areas, while 39.2% had varying types of vegetation. Wetlands accounted for 3% of the total area. Comparatively, by the final year (2020), there was a decrease in

no-vegetation areas to 54.4% and an increase in vegetation areas to 42.5%. Wetlands slightly increased from 3% to 3.2% during this period (Figure 3a–e).

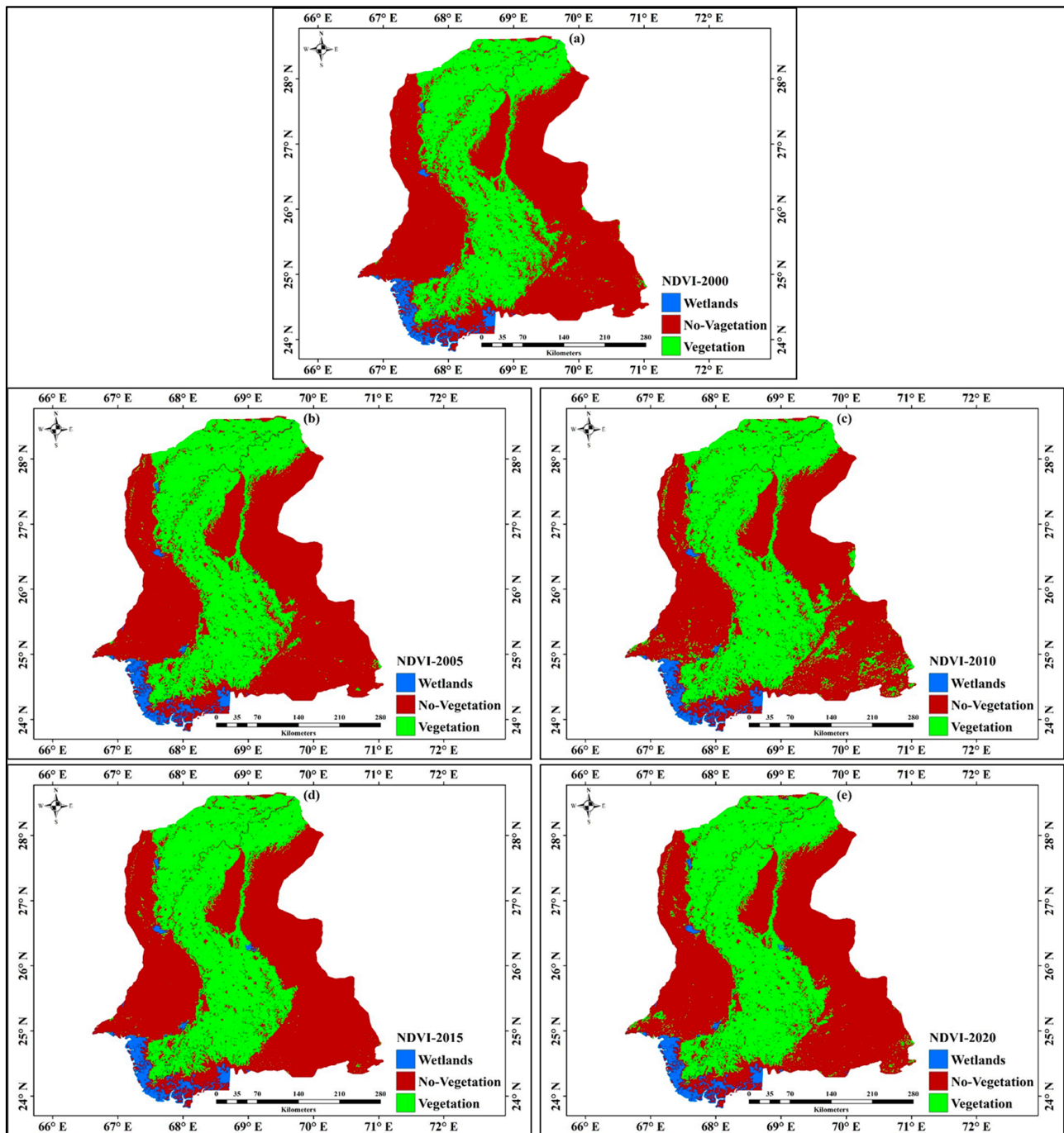


Figure 3. Categorized vegetation cover over Sindh province: (a) NDVI-2000; (b) NDVI-2005; (c) NDVI-2010; (d) NDVI-2015; (e) NDVI-2020.

The NDWI analysis involved two threshold values to classify land as either water or no water. In 2000, 97% of the land area was classified as having no water, with the remaining 2.2% being designated as wetlands. This represented a slight decrease from the previous year, where 96.8% of the land was classified as having no water. These results are depicted in Figure 4a–e.

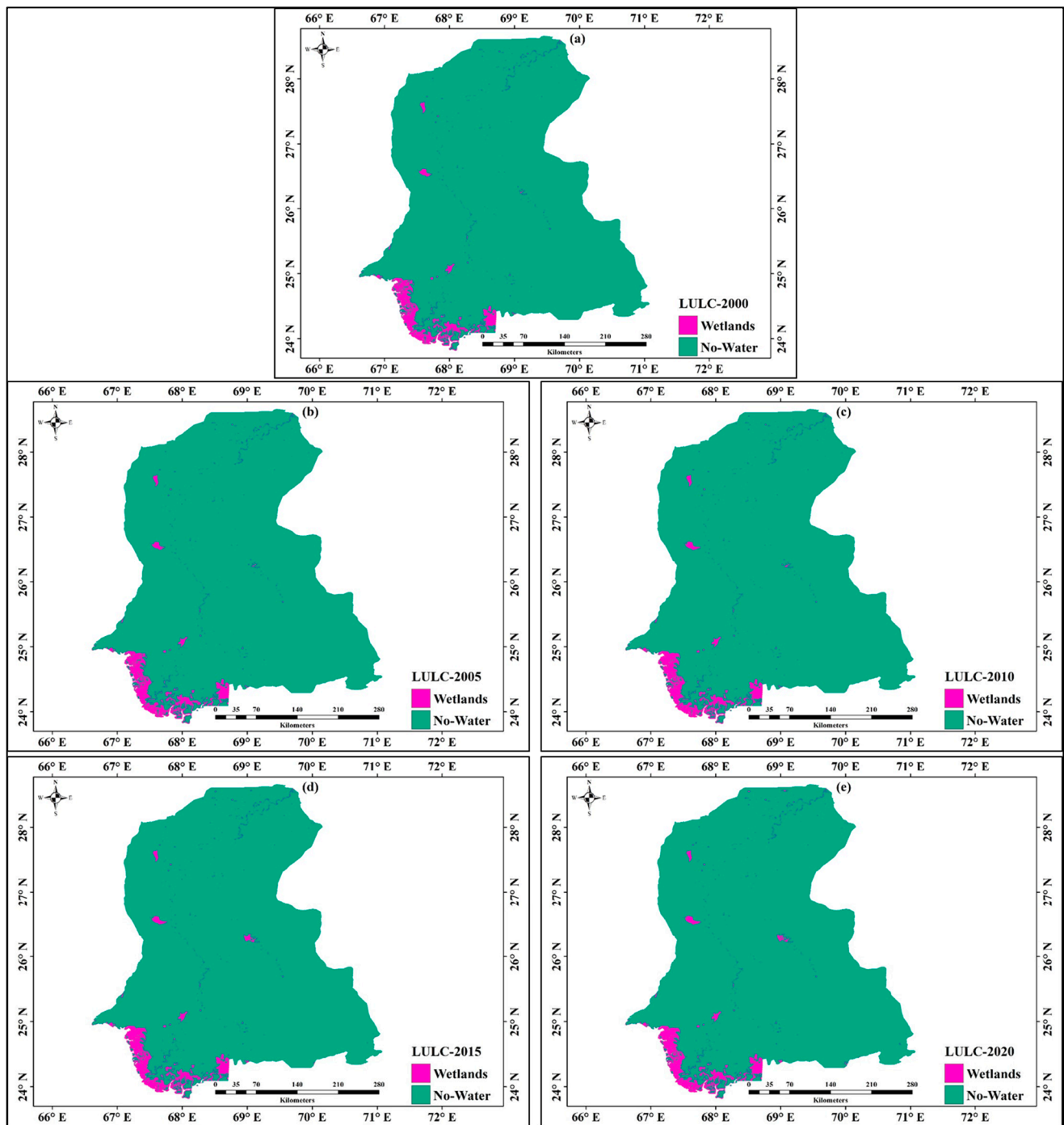


Figure 4. Categorized wetlands and no-water cover over Sindh province: (a) 2000; (b) 2005; (c) 2010; (d) 2015; (e) 2020.

Similarly, the NDBI threshold values were used to classify areas as water, no built-up, and built-up areas. In the base year (2000), 94% of the study area was categorized as no built-up areas, followed by 3% wetlands and approximately 1% built-up areas. Contrasting these results with those of the year 2020, an increase in no built-up areas to 95.2% and an increase in built-up areas to 1.7% were noted, alongside a minor rise in wetlands from 3% to 3.2% (Figure 5a–e).

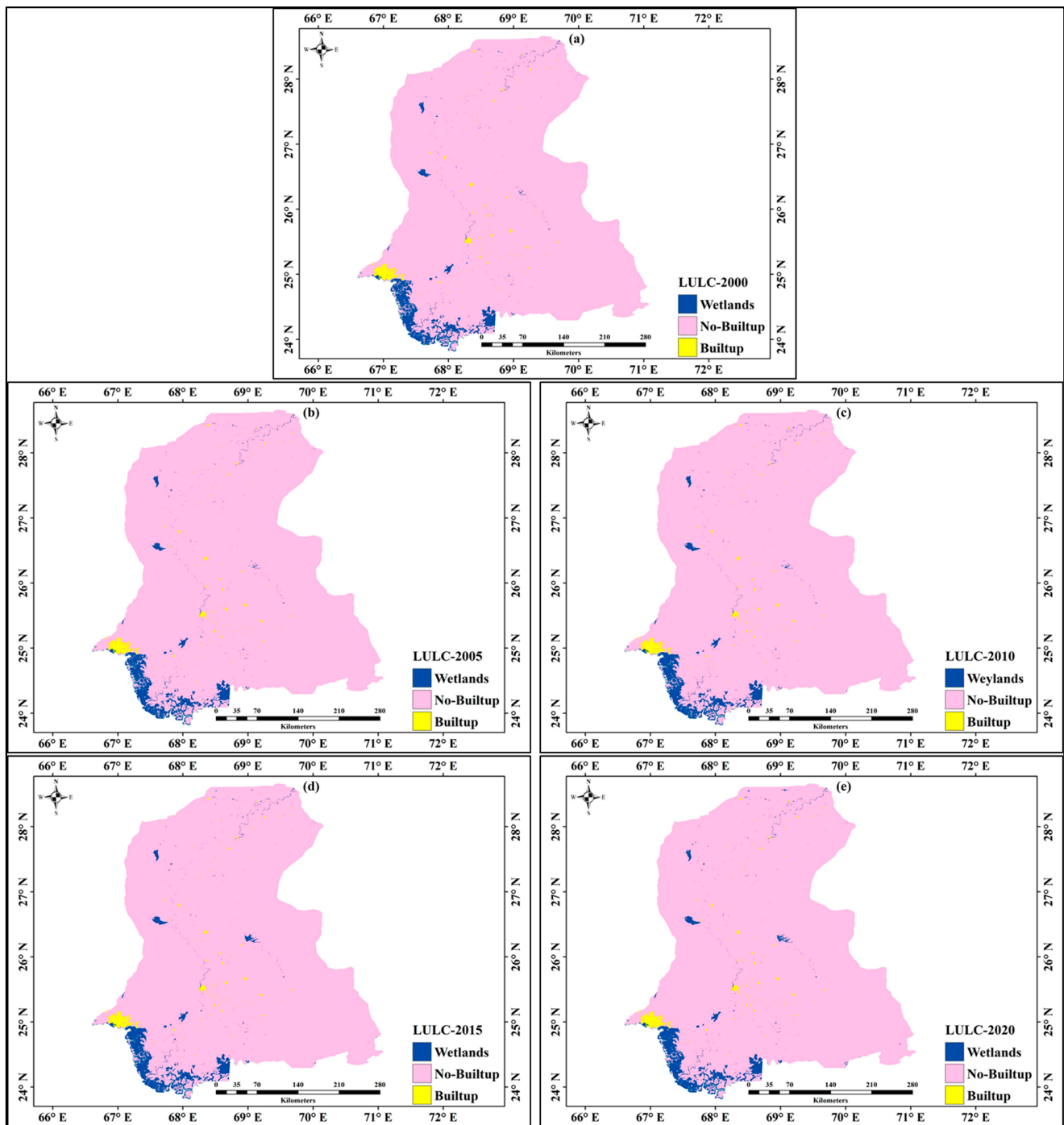


Figure 5. Categorized wetlands, built-up, and no-built-up areas over Sindh province: (a) 2000; (b) 2005; (c) 2010; (d) 2015; (e) 2020.

3.3. Accuracy Assessment of the Classified Images

Validating classified LULC maps is essential for ensuring their alignment with ground truthing. Therefore, we conducted five accuracy assessments as outlined in the methodology section. The detailed accuracy results of LULC classification and ground truthing evaluation are presented in (Table 3). The kappa coefficient for each year exceeded 0.90, except for 2010. Overall accuracy surpassed 90% for all years, except 2010, where the lowest and highest user's and producer's accuracies were recorded for barren lands and wetlands (78.7%, 79.2%; 98.3%, 98%, respectively). The precision levels for all other categories exceeded 90%. Consistently, water bodies received the most accurate classification across

almost every year, closely followed by forests. Despite covering less than 1% of the land area, forests exhibited remarkable accuracy. However, barren land in 2010 displayed lower accuracy compared with other LULC types. One potential explanation is that the disastrous flood event in 2010 significantly affected most barren lands and croplands, leading to decreased accuracy in their classification.

Table 3. Accuracy assessment of all classified LULC of Landsat images.

Accuracy Types	LULC Types	2000	2005	2010	2015	2020
User's accuracy (%)	Forest	95.7	93	96.2	92.6	90.7
	Wetland	99.8	97.1	98.3	100	100
	Cropland	78	81	88.3	86.3	91.5
	Built-up	97	90	89	90	88.5
	Barren land	99	97.3	78.7	97.3	79.6
	Rangeland	93	91	95.4	95	93.4
Producer's accuracy (%)	Forest	96	93	92.4	93	91.7
	Wetland	99.5	88.1	98	88.1	100
	Cropland	86	94.1	89.1	94.1	90.1
	Built-up	97	94.3	89	94.3	87.7
	Barren land	96	95	79.2	95	79.6
	Rangeland	89.6	90.5	95.4	90.5	93.4
Overall accuracy (%)		92.3	94.8	89.1	95.3	96.8
Kappa coefficient		90.4	92.1	88.4	93.5	94.5

3.4. LULC Change Dynamics across the Indus Delta

Regarding LULC trends over 20 years in the Indus Delta region, it is crucial to differentiate between the terms “position” and “rate of change”, because they have distinct perspectives on LULC dynamics. The analysis of LULC data enables us to illustrate both the trend and rate of transition and map the conversion dynamics between different land cover types at the pixel level. To capture transition adjustments that are challenging to quantify, we collected multi-seasonal satellite data over an extended period. This study presents its findings in the following sequence: first and second-order LULC patterns for the periods 2000–2010 and 2010–2020, respectively, along with the total LULC pattern for 2000–2020 to assess the overall change trend and transition. Then, we selected small temporal changes to measure the annual change rate, while larger orders were chosen to evaluate the long-term transition potential. Lastly, we utilized these long-term temporal changes, based on small temporal change intervals, to anticipate the future direction of possible LULC change transitions driven by predicted factors, assessing changes in Sindh Province, Pakistan.

3.4.1. First-Order LULC Pattern (2000–2010)

During the first-order LULC pattern analysis from 2000 to 2010, significant changes in land cover types were observed. Initially, barren land dominated the landscape, covering approximately 57.1% of the area, followed by rangeland at 16.2%, cropland at 22.8%, and wetlands categorized as mixed at 3%. By 2005, there was a noticeable decrease in barren land and rangeland, which decreased to 55.7% and 15.2%, respectively. Meanwhile, built-up areas and cropland collectively increased to nearly 28% of the total, while water bodies remained stable at 3%. This shift in land cover distribution indicates a relatively rapid conversion of barren land to other LULC types compared with forest and rangeland. Notably, approximately 17.1% of all LULC conversions by 2010 involved the transformation of barren land. There was a modest increase in built-up areas (0.3%) and a significant expansion of cropland (4.4% by 2020), accompanied by a decline in barren land (−5.4%). Figure 6a presents a thematic map illustrating the conversion changes between LULC types, highlighting transitions from one category to another. The analysis revealed that

approximately 89% of the area remained unchanged, while the remaining 11% experienced conversion between various LULC types. The most significant conversions occurred from barren land to rangeland (5.2%) and from rangeland to cropland (4.5%), indicating dynamic shifts in land use patterns over the study period.

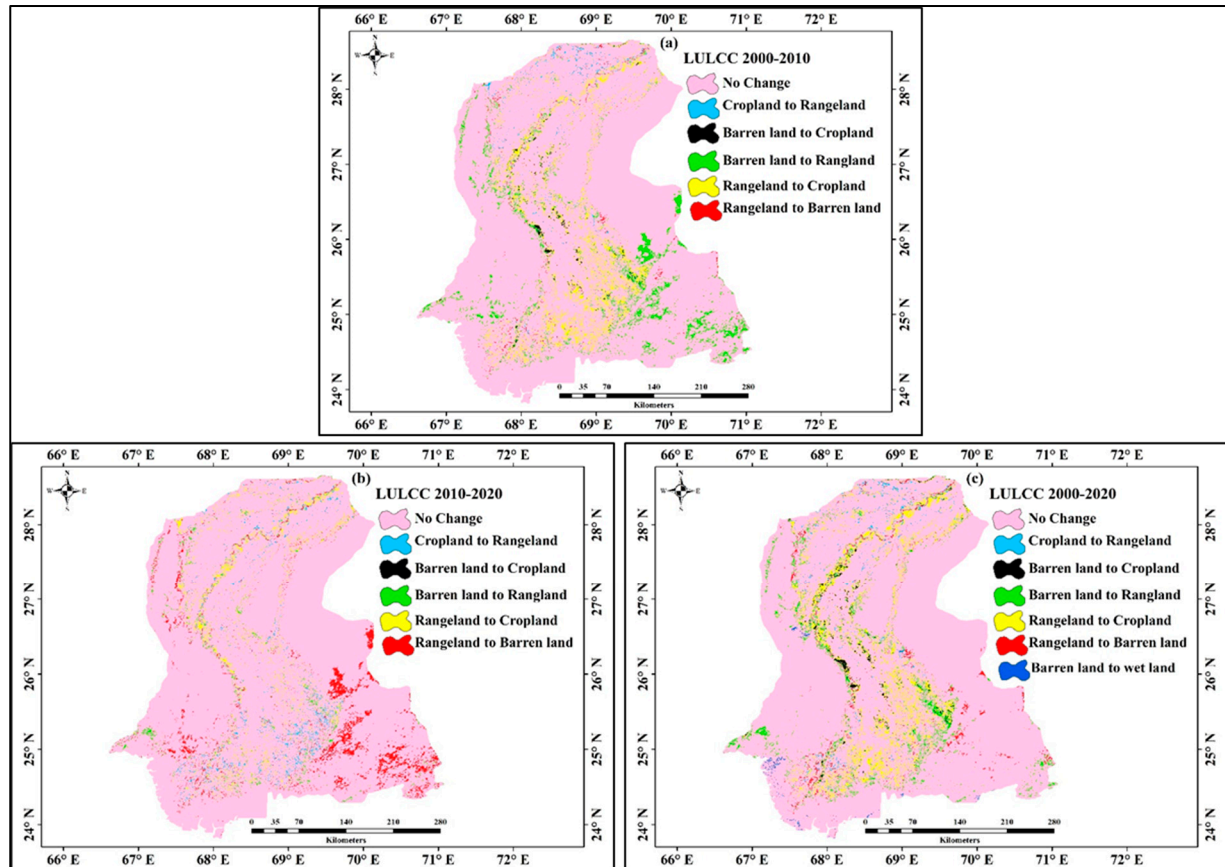


Figure 6. (a) First-order thematic conversion change map; (b) second-order thematic conversion change map; (c) the total-order thematic conversion change map.

3.4.2. Second-Order LULC Pattern (2010–2020)

In the analysis of second-order LULC patterns, similar trends as those observed in the first-order analysis were noted. However, a notable change was detected by 2015, as depicted in Figure 6b. Contrary to the continuous increase in cropland observed in previous years, barren land experienced a decrease of 0.6% and 2.7%, respectively, by 2020. During this period, there was a slight increase in forest and built-up areas by 0.2% and 0.7%, respectively, while the spatial extent of rangeland decreased compared with 2015. Overall, there was a 3.0% increase in total LULC types, while rangeland experienced a decline of the same magnitude. Figure 6b illustrates a thematic map of conversion changes, highlighting transitions from one LULC type to another. The analysis revealed that approximately 91.8% of the area remained unchanged, while the remaining 8.2% underwent conversion between various LULC types. By comparing with the study of Shah and Ai, 2024 [9] observed a slight increase in forests, built-up areas, and wetlands during this period, suggesting that the trend of increasing water bodies may be attributed to changes in rainfall patterns post-2010. This indicates the influence of climatic factors on the dynamics of land cover across the study area.

3.4.3. Total LULC Pattern (2000–2020)

Figure 6c provides an overview of the comprehensive LULC change pattern, explaining significant variations in both the volume and rate of change for each land cover type

across the region. Over two decades, there were notable shifts in land cover dynamics. The cropland area decreased by 4.4%, and rangeland saw a decline of 2.2%, while cropland itself increased to 5.4% of the total. Meanwhile, wetlands and forests remained relatively stable, with a modest increase of 0.2% and 0.1%, respectively. Additionally, there was a slight uptick of approximately 1% in built-up areas, indicating natural growth and urbanization trends over the past two decades. The analysis of conversion transitions reveals that roughly 89.5% of the area remained consistent, while the remaining 11.5% underwent conversion between all LULC groups. In the current scenario, there has been a significant shift in the pace of urban growth and rapid economic development, particularly in agriculture, over the last two decades, with 6.3% of the area undergoing significant changes. This emphasizes the dynamic nature of land cover and the complex relationship of various factors shaping the landscape over time.

3.5. Extraction of a Predictor Driver Responsible for LULCC

In our assessment of LULCC, we considered a variety of physical and climatic drivers, as well as static factors such as slope and elevation to capture topographical distinctions. To initiate this analysis, topographical features, including slope and elevation, wield considerable influence on landscape dynamics, shaping patterns of fragmentation and development. Steep slopes and elevated terrains often incur higher construction costs and pose challenges for development, contrasting with the more amenable conditions of flat and gently sloping areas. In our study area, encompassing the vast plains of the Indus and coastal and mangrove regions, topographic factors significantly influence LULCC. Slope maps depict gradients ranging from zero degrees in the southern coastal and mangrove zones to 39.6 degrees in the mountainous western regions (Figure 7a,b). Likewise, elevation maps delineate the predominantly flat southwestern regions in contrast to the higher-altitude western areas. Notably, settled areas tend to cluster in regions characterized by gentle to moderate slopes, with Karachi, Pakistan's densely populated metropolis, being situated on the southwestern coast near the Indian Ocean.

Next, we delineated the watershed boundaries within the study area using DEM data. Utilizing the "Euclidean Distance" tool, we computed the distance from streams, with the study area boundary serving as a mask, as illustrated in Figure 7b. Recognizing the critical importance of surface water, we acknowledge the potential risks posed by its overexploitation, particularly in the context of unplanned urbanization and intensive agricultural practices. Research emphasizes the reflective impact of these drivers on the natural flow patterns of surface water which, in turn, can significantly affect various ecosystem services. Therefore, we incorporated this driver into our predictive modeling for future LULC scenarios.

Road networks are another important parameter that facilitates human mobility and the transportation of goods, shaping landscape dynamics in urban and peri-urban areas. The density of roads often correlates with urban congestion and vegetation loss, reflecting the landscape's transformation. In our study, we focused on assessing LULCC trends by considering the distance from roads rather than road density, as depicted in Figure 7c. To ensure that the study's focus remained manageable, we included main expressways, primary roads, and highways in our analysis, omitting secondary and tertiary routes.

Our analysis of LULCC drew heavily upon extensive datasets of annual average rainfall and temperature spanning two decades (2000–2020), which is visually depicted in Figure 7e,f. Climate dynamics are paramount in understanding LULCCs, particularly in our study region, which has observed distinguished climate events such as floods in 2010, 2014, 2022, and 2023, alongside a severe heatwave in 2015. The rainfall distribution showcases significant variability, ranging from 81 mm in the northern regions to a substantial 946 mm in the southern coastal areas. Similarly, the temperature patterns indicate prevailing temperatures of 28 °C across most of the study area, except for the western high-altitude regions, where temperatures dip to 11.4 °C.

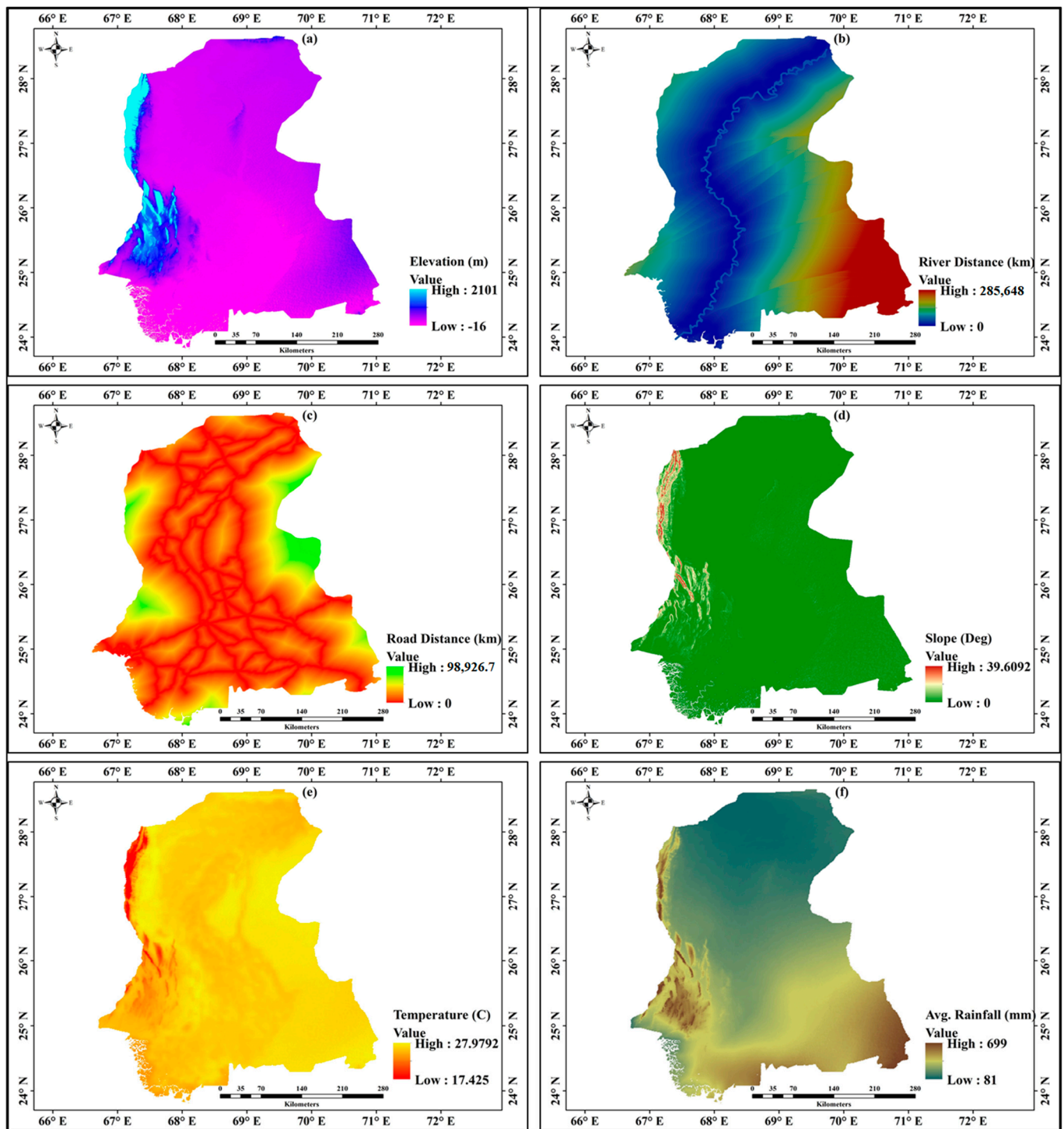


Figure 7. (a) Elevation; (b) river distance; (c) road distance; (d) slope; (e) avg. temperature; (f) avg. rainfall.

3.6. Future LULC Scenario Simulation

In this section, to forecast future scenarios with implications for policy, environmental conservation, and economic management, we employed a synergistic approach integrating Markov chain (MC) and cellular automata (CA) techniques. This combined methodology allows for the assessment of both static and dynamic shifts in LULCCs. To construct a transformation potential matrix, we utilized two sets of variables: dependent variables representing the current LULC and five independent predictor drivers, including distance to roads, distance to watershed, rainfall, temperature, slope, and elevation. By employing a

random sampling method, we configured the model's parameters, setting the maximum iteration and neighborhood pixels to 1000 and 9 cells, respectively, resulting in a kernel smoothing of 3×3 cells. Utilizing the MOLUSCE tool within QGIS 3.38 software, we modeled the transfer potential matrix using the artificial neural network (ANN) technique to simulate future LULC maps. To validate the accuracy of our model in forecasting LULC shifts, we compared the simulated LULC maps for the year 2020 with projected maps for the same year in the study area (Figure 8a). Validation was conducted using the QGIS-MOLUSCE validation module, measuring the overall kappa coefficients and percent correctness between classified and projected LULC maps for 2020 (Figure 8b). The results indicated an overall kappa coefficient of 0.886 and a 92.6% percent correctness (Figure 9), affirming the reliability and accuracy of our predictive model. Similarly, we further projected LULC maps for 2025 and 2030 (Figure 8c,d). Our findings emphasize the efficacy of MC-CA models, augmented by the ANN technique, as a robust tool for forecasting future LULC changes. This approach holds particular relevance for informing policies about land use and environmental conservation.

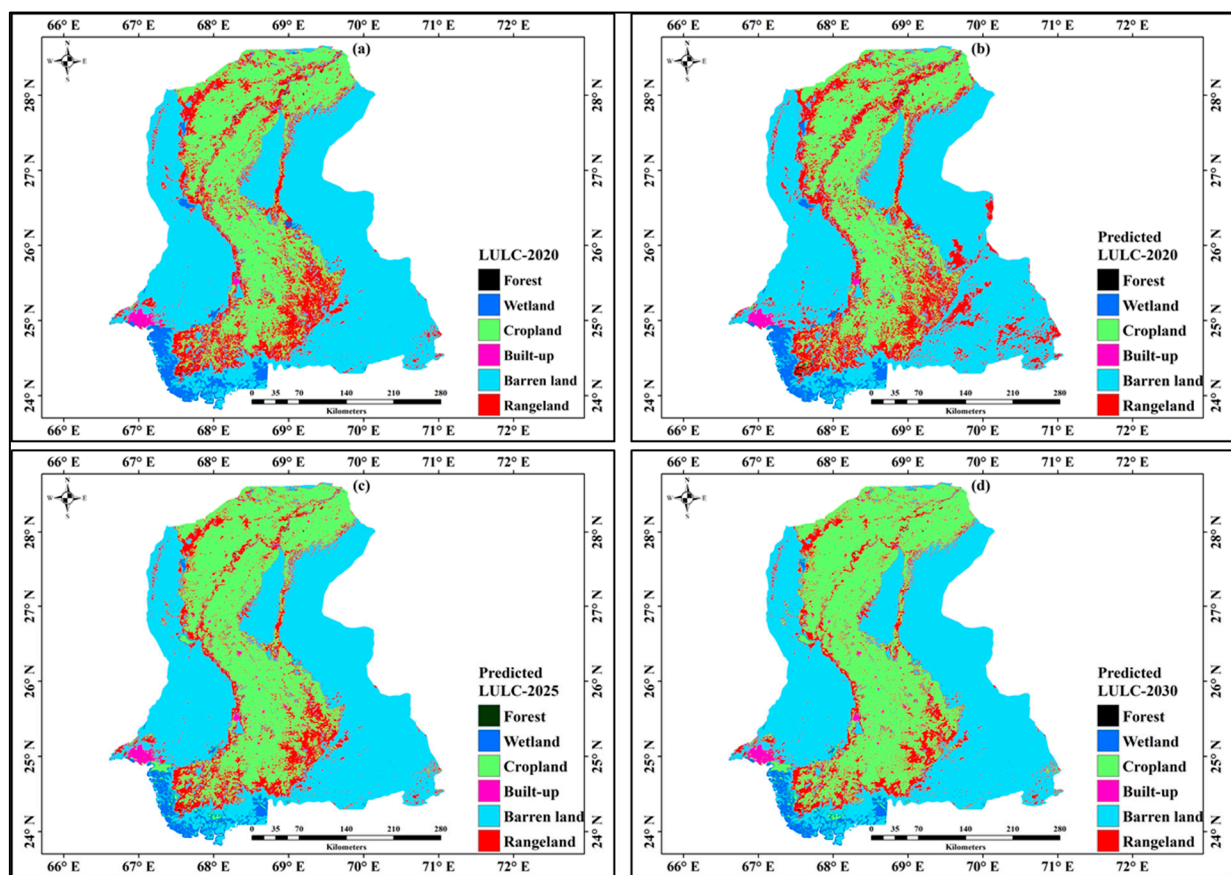


Figure 8. LULC prediction: (a) LULC-2020; (b) predicted LULC-2020; (c) predicted LULC-2025; (d) predicted LULC-2030.

The anticipation of LULC scenarios hinges upon past shifts in LULC and the predictor drivers. Prior research has delineated two simulated outcomes: soft prediction and hard prediction. While hard prediction sometimes amplifies results by exerting excessive pressure on predicted drivers, leading to the domination of all drivers over a particular contribution, soft prediction mitigates this tendency. To maintain fidelity and prevent exaggeration, this research exclusively incorporated soft prediction for the year 2030. Soft prediction for 2030 was derived from a division into two 10-year temporal spans, aligning with the overarching LULCC timeframe. Thus, the term “soft prediction” denotes the simulated results for 2025, while hard prediction represents the projection for 2030. The

forecasted LULC maps are depicted in Figure 8c,d. The simulated results for 2025 indicate a moderate reduction in wetlands (0.8%) and a substantial decline in rangeland and barren land, by -3.1% and -1.6% , respectively. In contrast, there is a notable increase of 5% in cropland and 0.3% in built-up areas. Similarly, the 2030 projections depict a continuation of this trend, with a slowdown in the increasing cropland and decreasing rangeland trends. Nonetheless, a modest increase in built-up and forest areas at 0.7% and 0.2%, respectively, remained apparent (Figure 10).

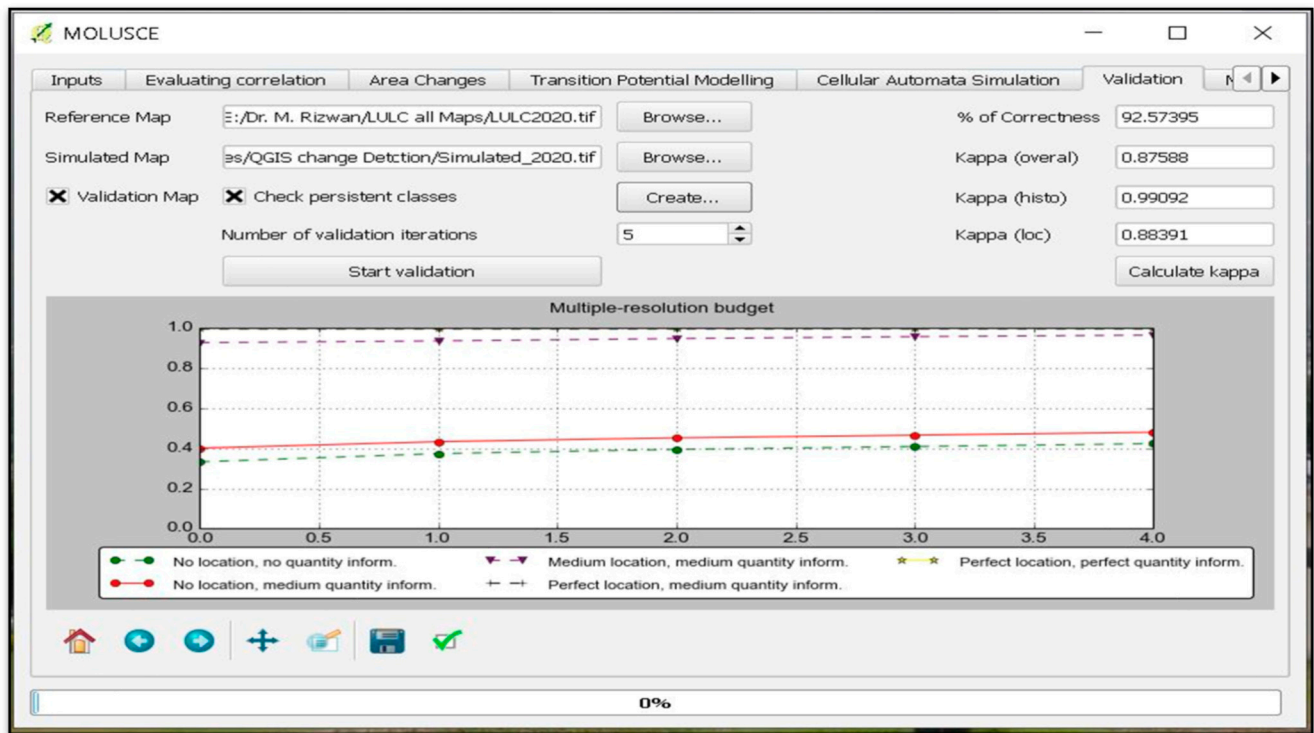


Figure 9. Cross-validation of the classified LULC map and simulated LULC map of 2020.

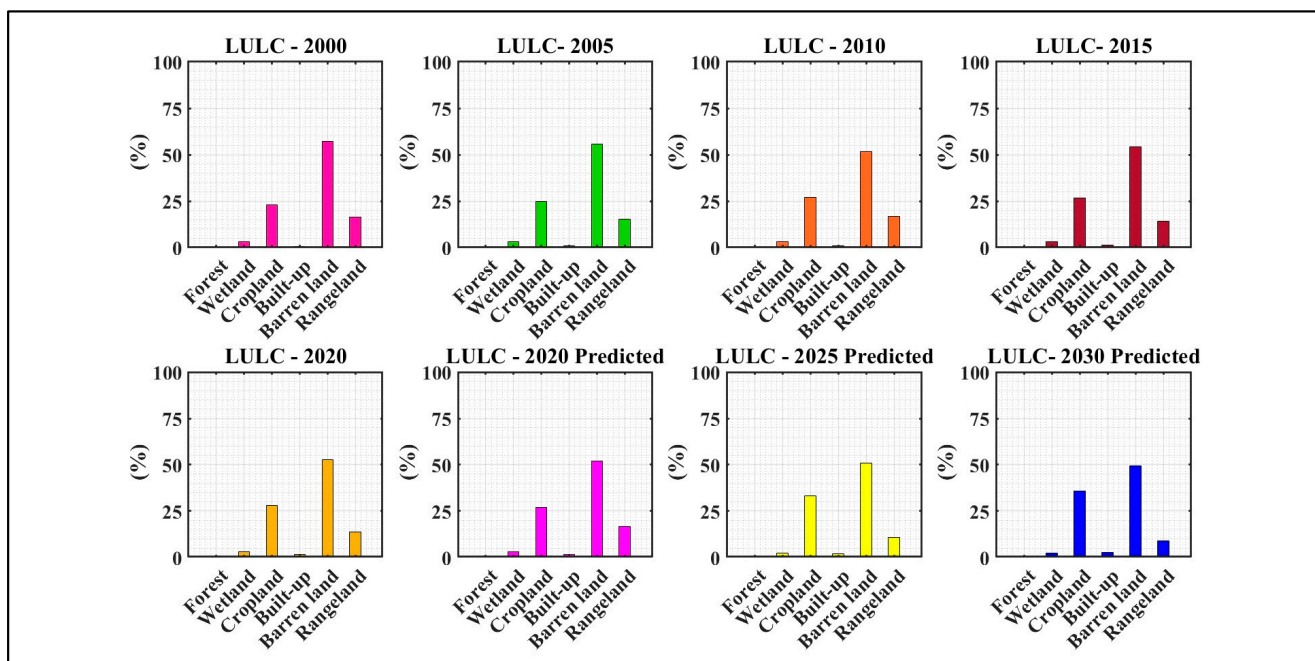


Figure 10. LULC change the trend pattern.

The predicted LULC maps demonstrate a moderate pace of conversion compared with the previous base years 2000, first-order 2000–2011, and second-order 2011–2020 change shifts. Nevertheless, the predicted drivers yield satisfactory outcomes, with water bodies experiencing modest increases in both orders. However, the substantial urban growth and economic development of near-surface water resources exert considerable pressure, which is projected to reduce these resources by 0.7% shortly around 2030. The heatmap of all LULCC analyses from 2000 to 2030 depicted in Figure 11 illustrates the change analysis for LULC types between different intervals from 2000 to 2030. Each cell indicates the rate of change for a specific LULC type over a five-year interval, with the following insights:

- Red color cells signify an increase in the percentage area of a specific LULC type over the interval, while blue cells indicate a decrease. The color intensity reflects the magnitude of the change.
- Forest areas witnessed a significant decrease between 2010 and 2015, succeeded by a gradual recovery, signifying deforestation followed by reforestation or natural regrowth efforts.
- Wetlands exhibit slight fluctuations with a general trend towards decrease, notably towards 2030, highlighting pressures on these ecosystems.
- Cropland consistently registers an increase across all intervals, reflecting ongoing agricultural expansion.
- Built-up areas show a steady rise, indicating urban growth.
- Barren land and rangeland display decreases in several intervals, suggesting land conversion to other uses or degradation.

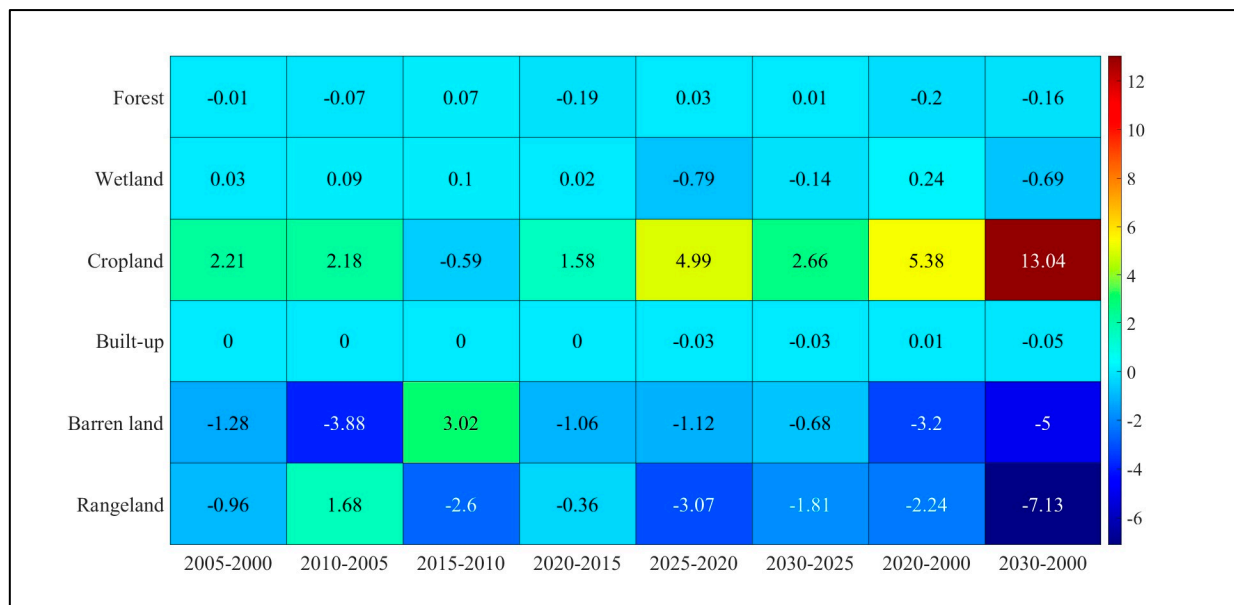


Figure 11. Heatmap illustrating the change analysis for LULC types from 2000 to 2030.

Additionally, accessibility near major roads and highways plays an important role in economic development and urban growth. By the near future (2030), it is predicted that all settled cities and towns will expand their urban fabric at the expense of barren land and rangeland, encompassing almost 13.1% of the land area. To meet the demands of the growing population trend, future simulation results suggest that cropland will cover up to 35.8% with an overall increment of 13% compared with the base layer. To support this assertion, a correlation heatmap demonstrates the correlation coefficients between different LULC types (Figure 11), indicating potential interactions between them, such as the relationship between agricultural expansion (cropland increase) and natural habitat reduction (forest and wetland decrease).

3.7. Special Emphasis on Mangrove Ecosystems

Mangroves hold special significance within the coastal ecosystem of the Indus Delta region in Sindh province, Pakistan, and provide a multitude of environmental, economic, and social benefits. Mainly situated along the southern coast of Sindh province, these mangrove forests form one of the world's largest arid-climate mangrove ecosystems. However, they face significant threats, including pollution from agricultural and industrial sources, as well as the adverse effects of climate change, such as sea level rise and increased salinity. Despite these challenges, recent studies indicate a slight growth in wetlands from 2.95% in 2000 to 3.19% in 2020, suggesting a modest increase in water bodies that could support mangroves. However, this positive trend is overshadowed by the significant expansion of cropland, which surged from 22.76% in 2000 to 28.14% in 2020, reflecting agricultural expansion at the expense of natural landscapes. This expansion threatens water runoff and sedimentation patterns crucial for mangrove health. Similarly, the growth of built-up areas from 0.71% in 2000 to 1.66% in 2020 signifies urban and infrastructural development that poses risks of habitat fragmentation and increased pollution. The increase in cropland areas suggests heightened freshwater diversion for irrigation, potentially intensifying agricultural runoff-carrying pollutants and sediments detrimental to mangrove habitats. Furthermore, the expansion of built-up areas contributes to pollution and disrupts natural water flow, further jeopardizing mangrove health and extent. Predictive models forecast continued changes by 2025 and 2030, including a projected increase in cropland to 35.80% by 2030, exacerbating pressures from agricultural runoff and water diversion. Concurrently, built-up areas are anticipated to grow to 2.7% by 2030, heightening habitat fragmentation and pollution. The decline in barren land to 49.5% and rangeland to 9.03% by 2030 indicates ongoing land use conversion that could further impact natural habitats, including mangroves. These predicted changes draw attention to the urgent need for sustainable land management practices aimed at protecting and restoring mangrove ecosystems. Addressing the projected increases in cropland and built-up areas, alongside the decrease in natural habitats, necessitates integrated coastal zone management strategies that balance development with mangrove forest conservation. Such measures are crucial for safeguarding the invaluable ecological services provided by mangroves and ensuring the resilience of coastal ecosystems in the face of ongoing environmental challenges.

4. Discussion

Research on the impacts of LULCC dynamics across the Indus Delta of Pakistan is vital due to its implications for ecosystem services, biodiversity, and human livelihoods. Our study reveals significant transformations in land cover patterns over the past two decades, characterized by a decline in natural vegetation and a concurrent increase in agricultural and urban areas. These findings align with global trends in LULCC, where urbanization and agricultural intensification are common drivers of landscape change [69–72].

In 2000, barren land dominated the landscape, covering 57.1% of the area, indicative of extensive regions with minimal vegetation. By 2020, this proportion had decreased to 52.7%, reflecting a transition towards more vegetated and developed land uses. This shift is consistent with observations in other regions where natural land cover is converted to meet the demands of growing populations and economic development [73]. The expansion of cropland from 22.8% in 2000 to 28.1% in 2020 enhances food security but raises concerns about water use efficiency, soil degradation, and biodiversity loss [74]. Similarly, urban areas expanded from 0.7% to 1.7% during the same period, illustrating the challenge of balancing developmental imperatives with environmental sustainability [75].

Changes in wetland areas, though modestly changing from 3% to 3.2% between 2000 and 2020, have significant implications for water resources and biodiversity conservation. Wetlands are crucial for water filtration, flood regulation, and habitat provision, and their conservation is critical amidst growing development pressures and climate change impacts [76]. Our analysis of LULCC trends across the Indus Delta region reveals a troubling pattern of diminishing natural habitats driven by agricultural expansion, urbanization,

and infrastructure development. While these trends align with broader economic development goals, they necessitate proactive management strategies to ensure the long-term sustainability of ecosystems and the wellbeing of local communities.

The accuracy of our LULC classification, with a kappa coefficient consistently exceeding 0.90 for most years except 2010, highlights the reliability of our findings. High classification accuracies for key land cover types such as forests and wetlands validate our analysis and provide insights into shifting land use patterns with implications for ecosystem services and biodiversity conservation [77]. Comparison with global and regional trends highlights both similarities and unique characteristics in LULCC across Sindh Province. Urbanization and agricultural intensification pressures are evident, reflecting global trends; however, local environmental conditions and socio-political factors significantly influence the extent and impacts of these changes.

The observed changes in LULCC, particularly the reduction in natural vegetation and the expansion of agricultural and urban areas underscore the need for sustainable land management practices. The conversion of forests and grasslands into agricultural land threatens biodiversity and compromises ecosystem services crucial for regulating climate, controlling floods, and maintaining water quality [78]. Sustainable development in the region requires effective land use policies that balance economic growth with environmental conservation. Predictive analyses indicate future challenges, with projected increases in urban and agricultural areas posing risks to environmental sustainability [79]. These forecasts emphasize the importance of integrating LULCC predictions into environmental policies and land use planning initiatives to promote sustainable development, preserve critical ecosystems, and enhance resilience against environmental challenges [80]. Lastly, our study highlights the complex dynamics of LULCC in the Indus Delta region and emphasizes the urgent need for integrated approaches to land use management. Addressing these challenges requires collaborative efforts among policymakers, researchers, and local communities to safeguard natural resources and ensure sustainable development pathways for the future [81].

5. Conclusions

This study consisted of the unique conditions of the coastal landscape in the Indus Delta, ensuring sensitivity to its specific challenges [82]. A key aspect of this adaptation involves enhancing the mapping and analysis of coastal features, particularly mangroves. These features necessitate specialized RS techniques due to their distinct spectral signatures and the dynamic nature of coastal zones [83]. For example, specific algorithms were employed in this study to differentiate between water and vegetation in mangrove areas, which is crucial for accurate land cover classification in delta regions. Addressing the high spatial variability and temporal dynamics inherent in coastal areas is another significant adaptation. The study employed a time-series analysis to capture annual changes in coastal land cover, which were more pronounced compared with inland areas. This involved analyzing satellite imagery from different seasons to comprehend the cyclical changes in coastal landscapes [84,85]. Therefore, consistent seasonal data were utilized to ensure coherence in the signature profile. Moreover, this study tackled challenges related to data inconsistency arising from factors such as cloud cover and atmospheric disturbances, which are more prevalent in deltaic plains. Advanced preprocessing techniques, including cloud masking and atmospheric correction, were employed to mitigate these issues and maintain the integrity of the satellite data, as detailed in the data collection section.

This study analyzed LULCC across Sindh Province, with a particular emphasis on the Indus Delta. By utilizing advanced RS techniques and MLAs, this study revealed the great implications of these changes on mangrove ecosystems. The primary aim was to measure the magnitude of transformations across diverse land cover types and devise strategies to reconcile ecological preservation with human development imperatives. The findings showed a noticeable propagation of agricultural and urban land use throughout the province, triggering an apparent change from natural vegetation cover. For instance, from

2000 to 2020, cropland expanded from 22.76% to 28.14%, while built-up areas surged from 0.71% to 1.66%, indicative of intensifying agricultural areas and urban sprawl. These shifts have substantially reconfigured the province's natural topography. Moreover, this study emphasized the vulnerability of mangrove forests in Sindh Province to an array of anthropogenic pressures. Despite a marginal uptick in wetland areas, rising from 2.95% to 3.19%, the encroachment of agricultural lands and built-up areas poses risks to ecosystems. Mangroves play a role in biodiversity conservation, coastal resilience, and climate variation, but they are challenged by the threat of decreasing deforestation, pollution, and the impacts of climate change. Projections indicate that the observed trends in land use change will persist, with further increments in cropland and urban areas being anticipated by 2030, impairing burdens on natural ecosystems, including mangroves. Hence, urgent action is imperative to implement sustainable land management practices. To tackle these challenges, targeted conservation programs aimed at conservation and rehabilitating mangrove ecosystems should be instituted. Additionally, this study underlines the importance of adopting sustainable agricultural practices to mitigate adverse environmental impacts and promote sustainable urban development. By emphasizing these approaches, we can address the challenges posed by rapid land use change while striving for environmental sustainability. This study lays the groundwork for future research on LULCC across the Indus Delta and similar contexts. Future investigations could delve into longitudinal studies monitoring LULCC impacts on ecosystems, examine socio-economic ramifications on local communities, and explore the role of land use and land cover in climate change mitigation and adaptation strategies, fostering resilient landscapes and communities.

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References

1. Anderson, J.R. *A Land Use and Land Cover Classification System for Use with Remote Sensor Data*; US Government Printing Office: Washington, DC, USA, 1976.
2. Adams, J.B.; Smith, M.O.; Gillespie, A.R. Imaging Spectroscopy: Interpretation based on Spectral Mixture Analysis. In *Remote Geochemical Analysis: Elemental and Mineralogical Composition*; Pieters, C.M., Englert, P., Eds.; Topics in Remote Sensing 4; Cambridge University Press: New York, NY, USA, 1993; pp. 145–166.
3. Asner, G.P.; Heidebrecht, K.B. Spectral unmixing of vegetation, soil and dry carbon cover in arid regions: Comparing multispectral and hyperspectral observations. *Int. J. Rem. Sens.* **2002**, *23*, 3939–3958. [\[CrossRef\]](#)
4. Waseem, S.; Khayyam, U. Loss of vegetative cover and increased land surface temperature: A case study of Islamabad, Pakistan. *J. Clean. Prod.* **2019**, *234*, 972–983. [\[CrossRef\]](#)
5. Khan, S.; Ullah, M.S.; Sun, T.; Rehman, A.U.R.; Chen, L. Land-use/land-cover changes and its contribution to urban heat island: A case study of Islamabad, Pakistan. *Sustainability* **2020**, *12*, 3861. [\[CrossRef\]](#)
6. Shahmohamadi, P.A.; Che-Ani, I.; Etesam, I.; Maulud, K.N.A.; Tawil, N. Healthy environment: The need to mitigate urban heat island effects on human health. *Procedia Eng.* **2011**, *20*, 61–70. [\[CrossRef\]](#)
7. Alqadhi, S.; Mallick, J.; Balha, A.; Bindajam, A.; Singh, C.K.; Hoa, P.V. Spatial and decadal prediction of land use/land cover using multi-layer perceptron-neural network (MLP-NN) algorithm for a semi-arid region of Asir, Saudi Arabia. *Earth. Sci. Inform.* **2021**, *14*, 1547–1562. [\[CrossRef\]](#)
8. Ahmed, B.; Wei, S.; Fu, Y.G.; Shabbir, M.; Nabi, G. Effects of floods policy in Pakistan and management issues: Case of District Dera-Ghazi Khan. *Int. J. Adv. Res.* **2014**, *2*, 967–974.

9. Shah, S.A.; Ai, S. Flood susceptibility mapping contributes to disaster risk reduction: A case study in Sindh, Pakistan. *Int. J. Disas. Risk Red.* **2024**, *108*, 104503. [\[CrossRef\]](#)
10. Faroughi, M.; Karimimoshaver, M.; Aram, F.; Solgi, E.; Mosavi, A.; Nabipour, N.; Chau, K.W. Computational modeling of land surface temperature using remote sensing data to investigate the spatial arrangement of buildings and energy consumption relationship. *Eng. Appl. Comput. Fluid Mech.* **2020**, *14*, 254–270. [\[CrossRef\]](#)
11. Khan, U.; Janjuhah, H.T.; Kontakiotis, G.; Rehman, A.; Zarkogiannis, S.D. Natural Processes and Anthropogenic Activity in the Indus River Sedimentary Environment in Pakistan: A Critical Review. *J. Mar. Sci. Eng.* **2021**, *9*, 1109. [\[CrossRef\]](#)
12. Altaf, M. The Major deserts of Pakistan—a review. *J. Wildl. Ecol.* **2022**, *6*, 78–92.
13. Ghafoor, G.Z.; Sharif, F.; Shahid, M.G.; Shahzad, L.; Rasheed, R.; Khan, A.U.H. Assessing the impact of land use land cover change on regulatory ecosystem services of subtropical scrub forest, Soan Valley Pakistan. *Sci. Rep.* **2022**, *12*, 10052. [\[CrossRef\]](#)
14. Waleed, M.; Sajjad, M.; Shazil, M.S. Urbanization-led land cover change impacts terrestrial carbon storage capacity: A high-resolution remote sensing-based nation-wide assessment in Pakistan (1990–2020). *Environ. Imp. Assess. Rev.* **2024**, *105*, 107396. [\[CrossRef\]](#)
15. Garcia, A.; Orenge, H.; Conesa, F.; Green, A.; Petrie, C. Remote sensing and historical morpho dynamics of alluvial plains. The 1909 Indus flood and the city of Dera Ghazi Khan (Province of Punjab, Pakistan). *Geosciences* **2018**, *9*, 21. [\[CrossRef\]](#)
16. Hussain, S.; Mubeen, M.; Ahmad, A.; Akram, W.; Hammad, H.M.; Ali, M.; Masood, N.; Amin, A.; Farid, H.U.; Sultana, S.R.; et al. Using GIS tools to detect the land use/land cover changes during forty years in Lodhran District of Pakistan. *Environ. Sci. Pollut. Res. Int.* **2020**, *27*, 39676–39692. [\[CrossRef\]](#)
17. Spalding, M. *World Atlas of Mangroves*; Routledge: London, UK, 2010. [\[CrossRef\]](#)
18. Mehmood, M.S.; Rehman, A.; Sajjad, M.; Song, J.; Zafar, Z.; Shiyan, Z.; Yaochen, Q. Evaluating land use/cover change associations with urban surface temperature via machine learning and spatial modeling: Past trends and future simulations in Dera Ghazi Khan, Pakistan. *Front. Ecol. Evol.* **2023**, *11*, 1115074. [\[CrossRef\]](#)
19. Saifullah, S. *Management of the Indus Delta Mangroves. Coastal Zone Management Imperative for Maritime Developing Nations*; Springer: Berlin/Heidelberg, Germany, 1997; pp. 333–346. [\[CrossRef\]](#)
20. Maurya, K.; Mahajan, S.; Chaube, N. Remote sensing techniques: Mapping and monitoring of mangrove ecosystem—A review. *Complex Intell. Syst.* **2021**, *7*, 2797–2818. [\[CrossRef\]](#)
21. Rodrigues, F.H.; Cerri, R.I.; de Andrade Kolya, A.; Veiga, V.M.; Reis, F.A.G.V. Comparison of vegetation indices and image classification methods for mangrove mapping at semi-detailed scale in southwest of Rio de Janeiro, Brazil. *Remote Sens. Appl. Soc. Environ.* **2023**, *30*, 100965. [\[CrossRef\]](#)
22. Purwanto, A.D.; Wikantika, K.; Deliar, A.; Darmawan, S. Decision tree and random forest classification algorithms for mangrove forest mapping in Sembilang National Park, Indonesia. *Remote Sens.* **2022**, *15*, 16. [\[CrossRef\]](#)
23. Rahmadi, R.; Koestoe, R.H.S. Comparison of the mangrove forest mapping algorithms in Kelabat Bay using random forest and support vector machines. *Int. J. Remote Sens. Earth Sci.* **2023**, *20*, 106–112.
24. Mukhtar, I.; Hannan, A. Constrains on mangrove forests and conservation projects in Pakistan. *J. Coast. Conserv.* **2012**, *16*, 51–62. [\[CrossRef\]](#)
25. Abbas, S.; Qamer, F.M.; Ali, G.; Tripathi, N.K.; Shehzad, K.; Saleem, R.; Gilani, H. An assessment of status and distribution of mangrove forest cover in Pakistan. *J. Biodiver. Environ. Sci.* **2013**, *3*, 64–78.
26. Chuvieco, E. *Fundamentals of Satellite Remote Sensing: An Environmental Approach*; CRC Press: Boca Raton, FL, USA, 2020. [\[CrossRef\]](#)
27. Purkis, S.J.; Klemas, V.V. *Remote Sensing, and Global Environmental Change*; John Wiley & Sons: Hoboken, NJ, USA, 2011. [\[CrossRef\]](#)
28. Rehman, A.; Song, J.; Haq, F.; Mahmood, S.; Ahamad, M.I.; Basharat, M.; Sajid, M.; Mehmood, M.S. Multi-hazard susceptibility assessment using the analytical hierarchy process and frequency ratio techniques in the Northwest Himalayas, Pakistan. *Remote Sens.* **2022**, *14*, 554. [\[CrossRef\]](#)
29. Thakur, S.; Mondal, P.; Ghosh, P.; Das, P.; De, T.K. A review of the application of multispectral remote sensing in the study of mangrove ecosystems with special emphasis on image processing techniques. *Spatial Inform. Res.* **2020**, *28*, 39–51. [\[CrossRef\]](#)
30. Karimzadeh, S.; Kamran, K.V.; Mahdavi, M. A multi-sensor satellite image classification for the detection of mangrove forests in Qeshm Island (Southern Iran). *Appl. Geomat.* **2023**, *15*, 177–188. [\[CrossRef\]](#)
31. Khan, R.; Li, H.; Basir, M.; Chen, Y.L.; Sajjad, M.M.; Haq, I.U.; Ullah, B.; Arif, M.; Hassan, W. Monitoring land use land cover changes and its impacts on land surface temperature over Mardan and Charsadda Districts, Khyber Pakhtunkhwa (KP), Pakistan. *Environ. Monit. Assess.* **2022**, *194*, 409. [\[CrossRef\]](#)
32. Prasad, P.; Loveson, V.J.; Chandra, P.; Kotha, M. Evaluation and comparison of the earth observing sensors in land cover/land use studies using machine learning algorithms. *Ecol. Informat.* **2022**, *68*, 101522. [\[CrossRef\]](#)
33. Reddy, G.O.; Kumar, N.; Singh, S. Remote sensing and GIS in mapping and monitoring of land degradation. *Geospat. Technol. Land Resour. Map. Monit. Manag.* **2018**, *21*, 401–424. [\[CrossRef\]](#)
34. Arab, S.T.; Islam, M.M.; Shamsuzzoha, M.; Alam, K.F.; Muhsin, N.; Noguchi, R.; Ahamed, T. A review of remote sensing applications in agriculture and forestry to establish big data analytics. In *Remote Sensing Application: Regional Perspectives in Agriculture and Forestry*; Springer: Singapore, 2022; pp. 1–24. [\[CrossRef\]](#)
35. Ghrefat, H.; Al Mutairi, Y.; ElAraby, H.; Galmed, M.; Mohamed, E. Using reflectance spectroscopy and Advanced Spaceborne Thermal Emission and Reflection Radiometer data to identify bauxite deposits in the vicinity of Az Zabirah, northern Saudi Arabia. *Arab. J. Geosci.* **2021**, *14*, 820. [\[CrossRef\]](#)

36. Varga-Szilay, Z.; Fetyko, K.G.; Szovenyi, G.; Pozsgai, G. Bridging biodiversity and gardening: Unravelling the interplay of socio-demographic factors, garden practices, and garden characteristics. *Urban For. Urban Green.* **2024**, *97*, 128367. [\[CrossRef\]](#)
37. Zomer, R.J.; Trabucco, A.; Bossio, D.A.; Verchot, L.V. Climate change mitigation: A spatial analysis of global land suitability for clean development mechanism afforestation and reforestation. *Agric. Ecosys. Environ.* **2008**, *126*, 67–80. [\[CrossRef\]](#)
38. Hussain, M.; Chen, D.; Cheng, A.; Wei, H.; Stanley, D. Change detection from remotely sensed images: From pixel-based to object-based approaches. *ISPRS J. Photogram. Remote Sens.* **2013**, *80*, 91–106. [\[CrossRef\]](#)
39. Hansen, M.C.; Potapov, P.V.; Moore, R.; Hancher, M.; Turubanova, S.A.; Tyukavina, A.; Thau, D.; Stehman, S.V.; Goetz, S.J.; Loveland, T.R. High-resolution global maps of 21st-century forest cover change. *Science* **2013**, *342*, 850–853. [\[CrossRef\]](#)
40. Pettorelli, N.; Laurance, W.F.; O'Brien, T.G.; Wegmann, M.; Nagendra, H.; Turner, W. Satellite remote sensing for applied ecologists: Opportunities and challenges. *J. Appl. Ecol.* **2014**, *51*, 839–848. [\[CrossRef\]](#)
41. Shah, S.A.; Kiran, M. Mann–Kendall test: Trend analysis of temperature, rainfall, and discharge of Ghotki feeder canal in district Ghotki, Sindh, Pakistan. *Environ. Ecosys. Sci.* **2021**, *5*, 137–142. [\[CrossRef\]](#)
42. Shah, S.A.; Kiran, M. A GIS-based technique analysis of land use and land cover change detection in taluka Mirpur Mathelo: A case study in district Ghotki, Pakistan. *Int. Adv. Res. Eng. J.* **2021**, *5*, 231–239. [\[CrossRef\]](#)
43. Demarquet, Q.; Rapinel, S.; Dufour, S.; Hubert-Moy, L. Long-term wetland monitoring using the Landsat archive: A Review. *Remote Sens.* **2023**, *15*, 820. [\[CrossRef\]](#)
44. Liu, Y.; Hou, X.; Li, X.; Song, B.; Wang, C. Assessing and predicting changes in ecosystem service values based on land use/cover change in the Bohai Rim coastal zone. *Ecol. Indica* **2020**, *56*, 106004. [\[CrossRef\]](#)
45. Shah, S.A.; Kiran, M.; Khurshid, T. Seepage losses measurement of Desert minor and development of gauge-discharge rating curve: A case study in District Ghotki, Sindh. *World Acad. J. Eng. Sci.* **2021**, *8*, 13–22. [\[CrossRef\]](#)
46. Khan, N.; Shah, S.J.; Rauf, T.; Zada, M.; Yukun, C.; Harbi, J. Socioeconomic impacts of the billion trees afforestation program in Khyber Pakhtunkhwa Province (KPK), Pakistan. *Forests* **2019**, *10*, 703. [\[CrossRef\]](#)
47. Kafy, A.A.; Shuvo, R.M.; Naim, M.N.H.; Sikdar, M.S.; Chowdhury, R.R.; Islam, M.A.; Sarker, M.H.S.; Khan, M.H.H.; Kona, M.A. Remote sensing approach to simulate the land use/land cover and seasonal land surface temperature change using machine learning algorithms in a fastest-growing megacity of Bangladesh. *Remote Sens. Appl. Soc. Environ.* **2020**, *21*, 100463. [\[CrossRef\]](#)
48. Tayyab, M. Management of surface water resources to mitigate the water stress in Karachi. In Proceedings of the International Conference on Hydrology and Water Resources (ICHWR-2021), Lahore, Pakistan, 25 March 2021.
49. Gedan, K.B.; Epanchin-Niell, R.; Qi, M. Rapid lands cover change in a submerging coastal county. *Wetlands* **2020**, *40*, 1717–1728. [\[CrossRef\]](#)
50. Zhang, F.; Yang, X. Improving land cover classification in an urbanized coastal area by random forests: The role of variable selection. *Remote Sens. Environ.* **2020**, *251*, 112105. [\[CrossRef\]](#)
51. Ai, J.; Zhang, C.; Chen, J.; Li, D. Mapping annual land use and land cover changes in the Yangtze estuary region using an object-based classification framework and Landsat time series data. *Sustainability* **2020**, *12*, 659. [\[CrossRef\]](#)
52. Chughtai, A.H.; Abbasi, h.; Karas, I.R. A review on change detection method and accuracy assessment for land use land cover. *Remote Sens. Appl. Soc. Environ.* **2021**, *22*, 100482. [\[CrossRef\]](#)
53. Tang, J.; Li, Y.; Cui, S.; Xu, L.; Ding, S.; Nie, W. Linking land-use change, landscape patterns, and ecosystem services in a coastal watershed of southeastern China. *Glob. Ecol. Conserv.* **2020**, *23*, e01177. [\[CrossRef\]](#)
54. Talukdar, S.; Singha, P.; Mahato, S.; Pal, S.; Liou, Y.A.; Rahman, A. Land-use land-cover classification by machine learning classifiers for satellite observations—A review. *Remote Sens.* **2020**, *12*, 1135. [\[CrossRef\]](#)
55. Rana, V.K.; Suryanarayana, T.M.V. Performance evaluation of MLE, RF, and SVM classification algorithms for watershed-scale land use/land cover mapping using sentinel 2 bands. *Remote Sens. Appl. Soc. Environ.* **2020**, *19*, 100351. [\[CrossRef\]](#)
56. Regasa, M.S.; Nones, M.; Adeba, D. A review on land use and land cover change in Ethiopian basins. *Land* **2021**, *10*, 585. [\[CrossRef\]](#)
57. Pal, S.; Ziaul, S. Detection of land use and land cover change and land surface temperature in English Bazar urban centre. *Egypt. J. Remote Sens. Space Sci.* **2017**, *20*, 125–145. [\[CrossRef\]](#)
58. Piyoosh, A.K.; Ghosh, S.K. Analysis of land use land cover change using a new and existing spectral index and its impact on normalized land surface temperature. *Geocarto Int.* **2022**, *37*, 2137–2159. [\[CrossRef\]](#)
59. Lemenkova, P.; Debeir, O. Computing Vegetation Indices from the Satellite Images Using GRASS GIS Scripts for Monitoring Mangrove Forests in the Coastal Landscapes of Niger Delta, Nigeria. *J. Mar. Sci. Eng.* **2023**, *11*, 871. [\[CrossRef\]](#)
60. Ahmad, A.; Khan, M.; Shah, S.H.H.; Kamran, M.; Wajid, S.A.; Amin, M.; Khan, A.; Arshad, M.N.; Cheema, M.J.M.; Saqid, Z.A. *Agro-Ecological Zones of Punjab, Pakistan*; FAO: Rome, Italy, 2019.
61. Platt, R.V.; Rapoza, L. An evaluation of an object-oriented paradigm for land use/land cover classification. *Prof. Geogr.* **2008**, *60*, 87–100. [\[CrossRef\]](#)
62. Tilahun, A.; Teferie, B. Accuracy assessment of land use land cover classification using Google Earth. *Am. J. Environ. Prot.* **2015**, *4*, 193–198. [\[CrossRef\]](#)
63. Rwanga, S.S.; Ndambuki, J.M. Accuracy assessment of land use/land cover classification using remote sensing and GIS. *Int. J. Geosci.* **2017**, *8*, 611. [\[CrossRef\]](#)
64. Kraemer, C.; Panda, S.S. Automating arc hydro for watershed delineation. In Proceedings of the 2009 Georgia Water Resources Conference, University of Georgia, Athens, GA, USA, 27–29 April 2009.

65. Rimal, B.; Zhang, L.; Keshtkar, H.; Wang, N.; Lin, Y. Monitoring and Modeling of Spatiotemporal Urban Expansion and Land-Use/Land-Cover Change Using Integrated Markov Chain Cellular Automata Model. *ISPRS Int. J. Geo-Inform.* **2017**, *6*, 288. [\[CrossRef\]](#)
66. Hamad, R.; Balzter, H.; Kolo, K. Predicting land use/land cover changes using a CA-Markov model under two different scenarios. *Sustainability* **2018**, *10*, 3421. [\[CrossRef\]](#)
67. Mansour, S.; Al-Belushi, M.; Al-Awadhi, T. Monitoring land use and land cover changes in the mountainous cities of Oman using GIS and CA-Markov modelling techniques. *Land Use Policy* **2020**, *91*, 104414. [\[CrossRef\]](#)
68. Ruben, G.B.; Zhang, K.; Dong, Z.; Xia, J. Analysis and projection of land-use/land-cover dynamics through scenario-based simulations using the CA-Markov model: A case study in Guanting reservoir basin, China. *Sustainability* **2020**, *12*, 3747. [\[CrossRef\]](#)
69. Wang, Y.; Sun, Y.; Cao, X.; Wang, Y.; Zhang, W.; Cheng, X. A review of regional and Global scale Land Use/Land Cover (LULC) mapping products generated from satellite remote sensing. *ISPRS J. Photogram. Remote Sens.* **2023**, *206*, 311–334. [\[CrossRef\]](#)
70. Zhai, J.; Xiao, C.; Feng, Z.; Liu, Y. Are there suitable global datasets for monitoring of land use and land cover in the tropics? Evidences from mainland Southeast Asia. *Glob. Planet. Chang.* **2023**, *229*, 104233. [\[CrossRef\]](#)
71. Seto, K.C.; Fragkias, M.; Guneralp, B.; Reilly, M.K. A Meta-analysis of global urban land expansion. *PLoS ONE* **2011**, *6*, e23777. [\[CrossRef\]](#)
72. Foley, J.A.; Defries, R.; Asner, G.P.; Barford, C.; Bonan, G.; Carpenter, S.R.; Chapin, F.S.; Coe, M.T.; Daily, G.C.; Gibbs, H.K.; et al. Global consequences of land use. *Science* **2005**, *309*, 570–574. [\[CrossRef\]](#)
73. Lambin, E.F.; Geist, H.J. *Land Use and Land Cover Change-Local Processes and Global Impacts*; Springer: Berlin/Heidelberg, Germany, 2006; p. 222.
74. Tilman, D.; Reich, P.B.; Knops, J.; Wedin, D.; Mielke, T.; Lehman, C. Diversity and productivity in a long-term grassland experiment. *Science* **2001**, *294*, 843–845. [\[CrossRef\]](#)
75. Grimm, N.B.; Faeth, S.H.; Golubiewski, N.E.; Redman, C.L.; Wu, J.; Bai, X.; Briggs, J.M. Global change and the ecology of cities. *Science* **2008**, *319*, 756–760. [\[CrossRef\]](#)
76. Mitsch, W.J.; Gosselink, J.G. The Value of Wetlands: Importance of scale and landscape setting. *Ecol. Econ.* **2000**, *35*, 25–33. [\[CrossRef\]](#)
77. Congalton, R.G. A Review of Assessing the Accuracy of Classifications of Remotely Sensed Data. *Remote Sens. Environ.* **1991**, *37*, 35–46. [\[CrossRef\]](#)
78. Millennium Ecosystem Assessment (MEA). *Ecosystems and Human Well-Being: Health Synthesis*; Island Press: Washington, DC, USA, 2005.
79. Verburg, P.H.; Kok, K.; Pontius, R.G.; Veldkamp, A. Modeling land-use and land-cover change. In *Land-Use and Land-Cover Change*; Lambin, E.F., Geist, H., Eds.; Global Change the IGBP Series; Springer: Berlin/Heidelberg, Germany, 2006. [\[CrossRef\]](#)
80. Turner, R.N.; Crisp, R.J.; Lambert, E. Imagining intergroup contact can improve intergroup attitudes. *Group Process. Intergroup Relat.* **2007**, *10*, 427–441. [\[CrossRef\]](#)
81. Ostrom, E. A general framework for analyzing sustainability of social-ecological systems. *Science* **2009**, *325*, 419–422. [\[CrossRef\]](#)
82. Ahmad, H.; Abdallah, M.; Jose, F.; Elzain, H.E.; Bhuyan, M.S.; Shoemaker, D.J.; Selvam, S. Evaluation and mapping of predicted future land use changes using hybrid models in a coastal area. *Ecol. Inform.* **2023**, *78*, 102324. [\[CrossRef\]](#)
83. O'Hara, C.; King, G.J.S.; Cartwright, J.H.; King, R.L. Multitemporal land use and land cover classification of urbanized areas within sensitive coastal environments. *IEEE Trans. Geosci. Remote Sens.* **2003**, *41*, 2005–2014. [\[CrossRef\]](#)
84. Abijith, D.; Saravanan, S. Assessment of land use and land cover change detection and prediction using remote sensing and CA Markov in the northern coastal districts of Tamil Nadu, India. *Environ. Sci. Pollut. Res.* **2021**, *29*, 86055–86067. [\[CrossRef\]](#)
85. Kuenzer, C.; Bluemel, A.; Gebhardt, S.; Quoc, T.V.; Dech, S. Remote sensing of mangrove ecosystems: A Review. *Remote Sens.* **2011**, *3*, 878–928. [\[CrossRef\]](#)

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